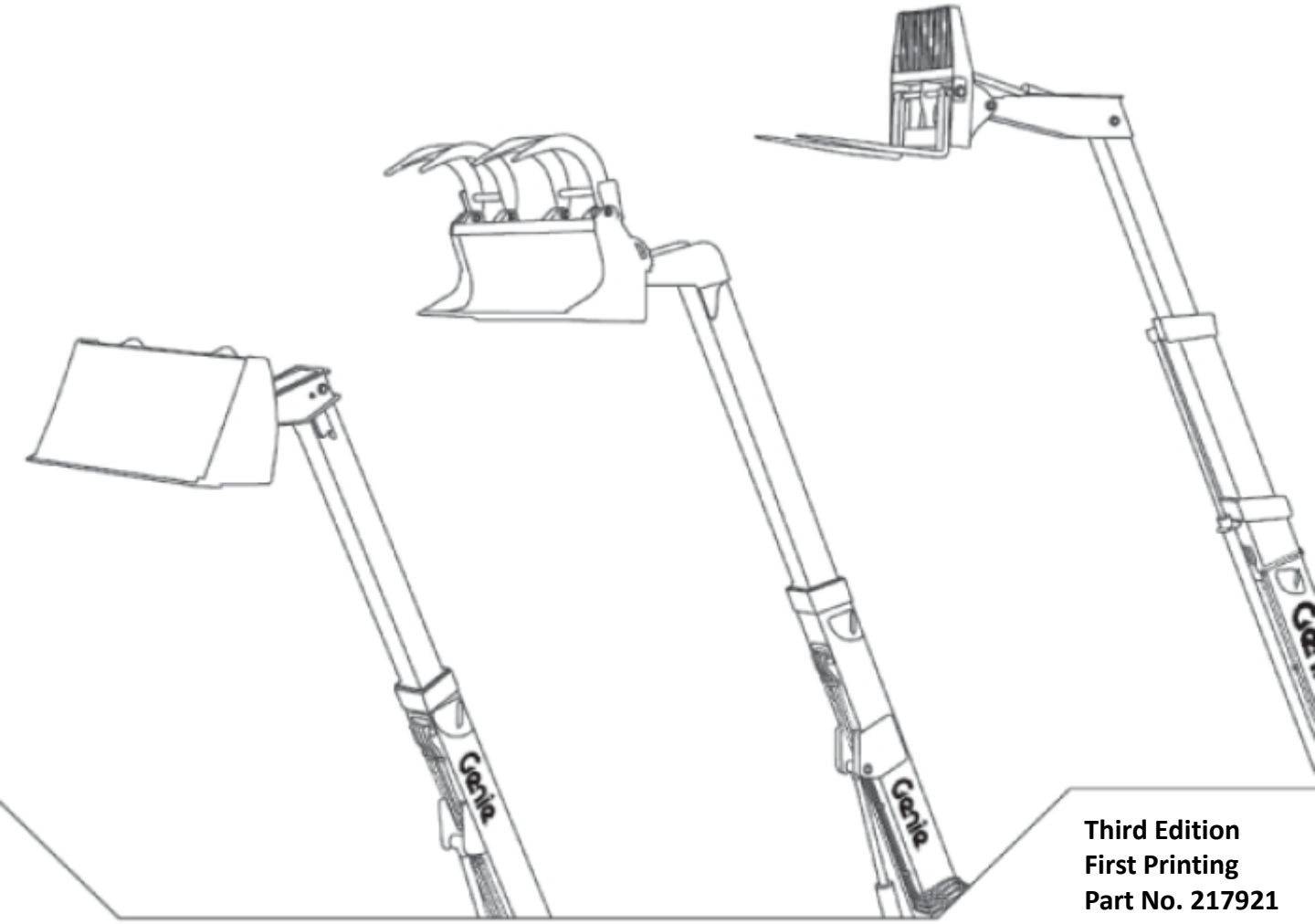
Genie®

LIFT PRO™

TELEHANDLER OPERATOR TRAINING

PARTICIPANTS GUIDE



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Responsibility

The employer must ensure that each powered industrial truck operator is competent and certified to operate a powered industrial truck safely, as demonstrated by the successful completion of the training and evaluation specified in the applicable standards, regulations and this training program.

The information contained in this training program is only provided to assist the employer and operator with complying with the applicable standards and regulations that apply to powered industrial truck training. Use of these materials will not result in automatic compliance with these standards (ANSI/ ITSDF B56.6, CSA B335) and regulations (local, state, provincial, or federal such as OHS or OSHA Rule 1910.178(l)).

This training program should only be delivered in its entirety. Additional machine or workplace specific knowledge and certification is the responsibility of the employer and the operator. Comply with all applicable job site, local, state, provincial or federal rules and regulations related to the proper safe use of the powered industrial trucks.

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Responsibility

Policy and Guidelines

Everyone associated with the manufacturing, ownership, sale, rental and operation of rough terrain forklift equipment should be committed to safe and proper operation. Failure to observe established operating practices and to comply with required responsibilities could result in death or serious injury.

Equipment users and operators can make a major contribution towards safety if they:

1. Comply with employer, job site and government rules.
2. Read, understand and follow the instructions in this guide and other manuals supplied with the equipment or any attachments.
3. Use good safe work practices in a commonsense way.
4. Be alert and free from the influences of alcohol, drugs or medications that might affect eyesight, hearing, reactions or judgment.
5. Only have qualified operators who are trained and authorized, directed by informed and knowledgeable supervisors, running the machine.

Read and understand the operator's manual and obey all safety rules and operating instructions before operating any rough terrain forklift equipment. The operator's manual must be considered a permanent part of your equipment and must always remain with the product.

If there is any information in this guide that is not clear or which you believe requires modification, please contact us.

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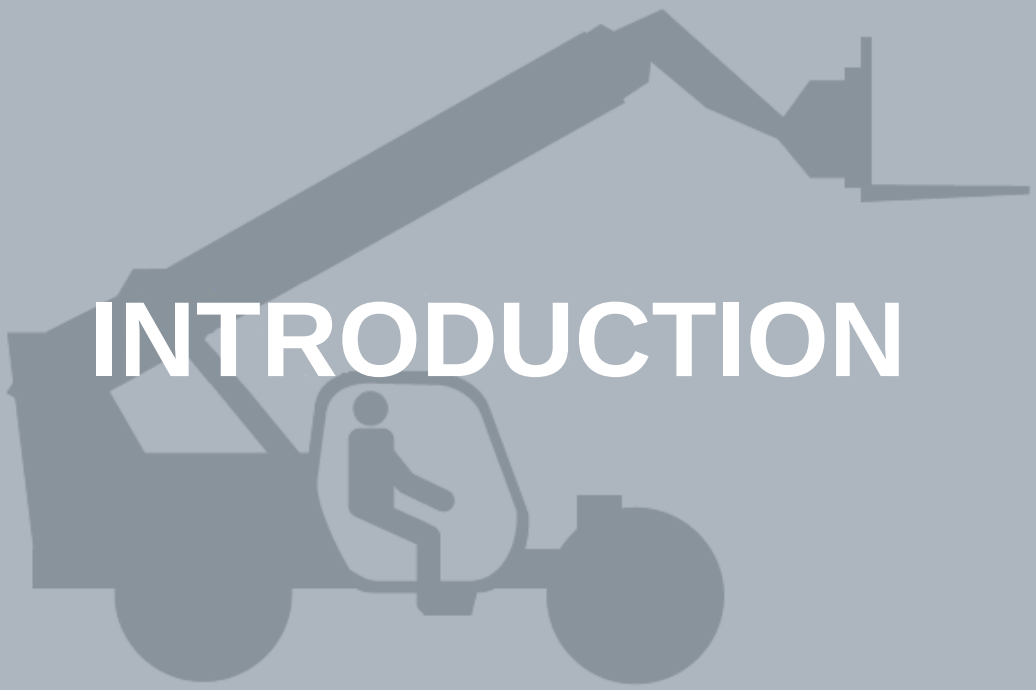

INTRODUCTION

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Overview

Scope

This training manual provides general training information for the purpose of certifying operators to operate a Class 7, variable reach rough terrain forklift truck, also known as a telehandler.

Definition

A *rough terrain forklift truck* is defined as a wheeled type truck designed primarily as a fork truck with a vertical mast and/or a pivoted boom, variable reach or of fixed length, which may be equipped with attachments. This truck is intended for operation on unimproved natural terrain as well as the disturbed terrain of construction sites. This definition excludes machines designed primarily for earth moving, such as loaders and dozers, even when their buckets and blades are replaced with forks, and machines designed primarily as over-the-road trucks equipped with lifting devices.

NOTE: Even though all Class 7 powered industrial trucks (which includes vertical mast forklifts, truck mounted forklifts and telehandlers) have similar working principles, this training program will focus only on telehandlers.

Course Curriculum

- **Theory**
  involves classroom discussions on subjects like regulations, company policies and procedures, operator's manuals, telehandlers features and safety equipment, and stability.

- **Hands-on training**
  involves the trainee (under the direct supervision of a trainer) using their physical ability to practice the safe operation of telehandlers. This includes the pre-operation inspection, workplace inspection, function test, start-up, and picking and placing of loads.

- **Knowledge Verification**
  involves evaluation to confirm the level of comprehension of the covered information and the hands-on training.

- **Practical Evaluation**
  involves the trainee's demonstration of proficiency in the safe operation of telehandlers and being measured against predetermined criteria.

Classroom Instruction

- ❑ Five Principles of Safe Operation
- ❑ Operation
- ❑ Options
- ❑ Maintenance
- ❑ Written Test

Hands-on Training

- ❑ Pre-operation Inspection
- ❑ Function Tests
- ❑ Workplace Inspection
- ❑ Demonstration of Proficiency

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Training Information

Qualification

Only _____and _____persons must be permitted to operate a telehandler (except for training purposes).

Definitions

By definition, the **user** is the person or persons who are in the position of directing operators to use the telehandler to perform work. Most commonly, this is the employer.

The person who actually operates the telehandler is the **operator**.

Training

The _____must verify that the following has been covered as part of operator's training.

1. Operators understand that safe operation is their responsibility.

2. Operators are knowledgeable of and observe the applicable safety rules and practices within the appropriate rules, regulations and standards that apply to the telehandler as well as the application.

Responsibility

1. Safe operation is the responsibility of the_____.

2. The operator must develop safe working habits and also be aware of hazardous conditions in order to protect themselves, other personnel, and property.

3. The operator must be familiar with the _____ and _____of all controls and instruments before operating the telehandler.

4. Before operating any telehandler, operators must _____ and be _____with the operator's manual for the particular telehandler being operated.

Operator's Manual

For telehandlers, and in many cases the attachment, manufacturers provide an operator's manual that contains key safety information that is related to the safe use of the equipment. As such it is a critical component, as a supplement, to this training program curriculum. The operator's manual should always be located_____the operator's cab.

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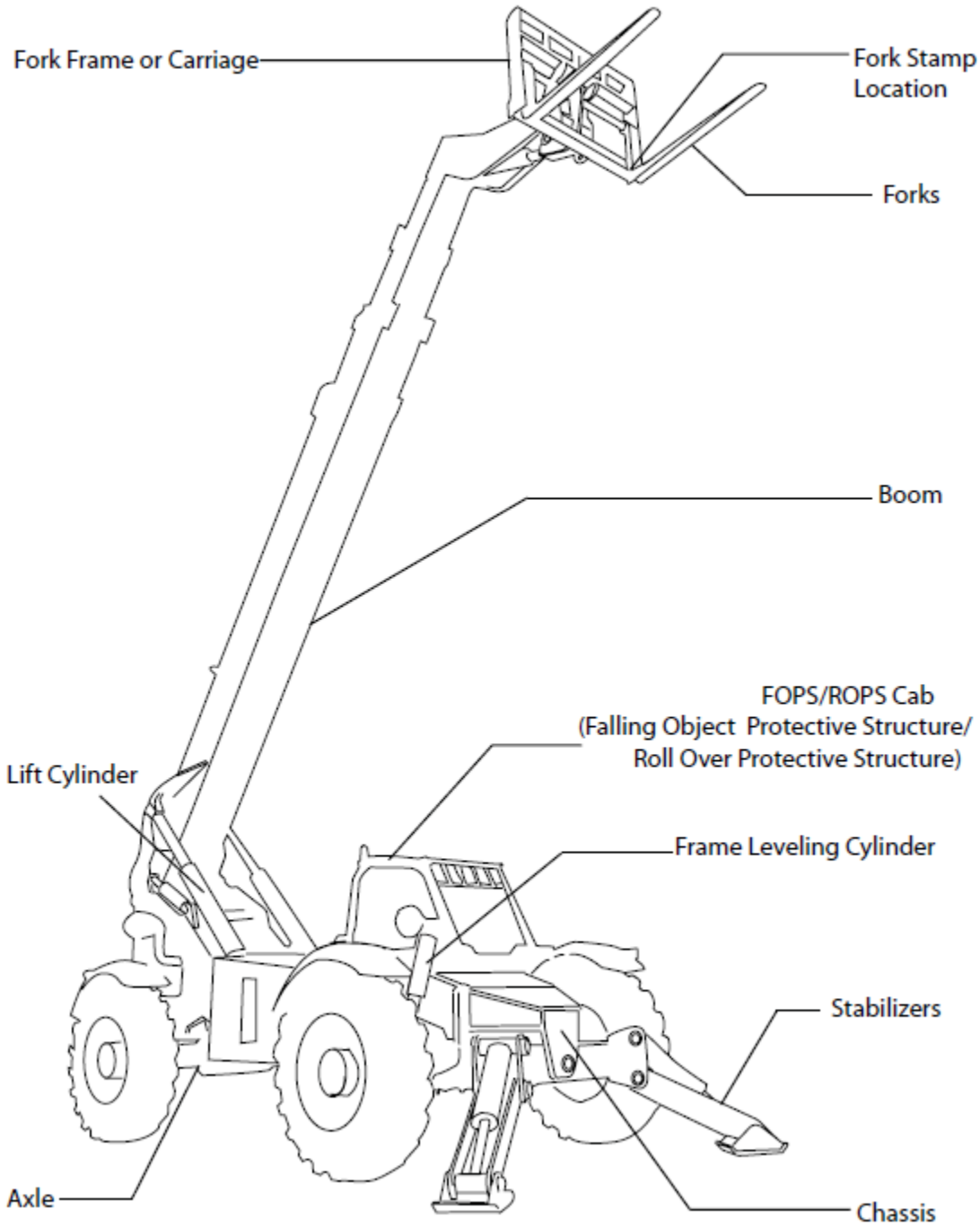
Legend



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Telehandler Features and Safety Equipment

Telehandlers are classified as _____ Variable Reach Rough Terrain Forklifts in the USA and Canada. They primarily serve construction and institutional markets where they are used for various applications.

Telehandlers incorporate various features and components that are essential to the safe operation of the machine. The operator should be familiar with and understand the proper use of all features of the machine.

Some telehandler features/components are listed below. We will cover many of these in detail in this program.

- Steering
- All controls and gauges
- Load charts and capacity plates
- Safety decals and markings
- Visual warning systems such as lights, rotating warning lights, flashing beacons
- Audible warning systems such as backup alarms, buzzers, and horns
- Operator restraint systems such as seat belts
- Fire extinguishers
- Mirrors
- Load backrest
- Stabilizers
- Rear axle lockout systems
- Auxiliary hydraulics
- Frame and boom level indicators
- Frame leveling

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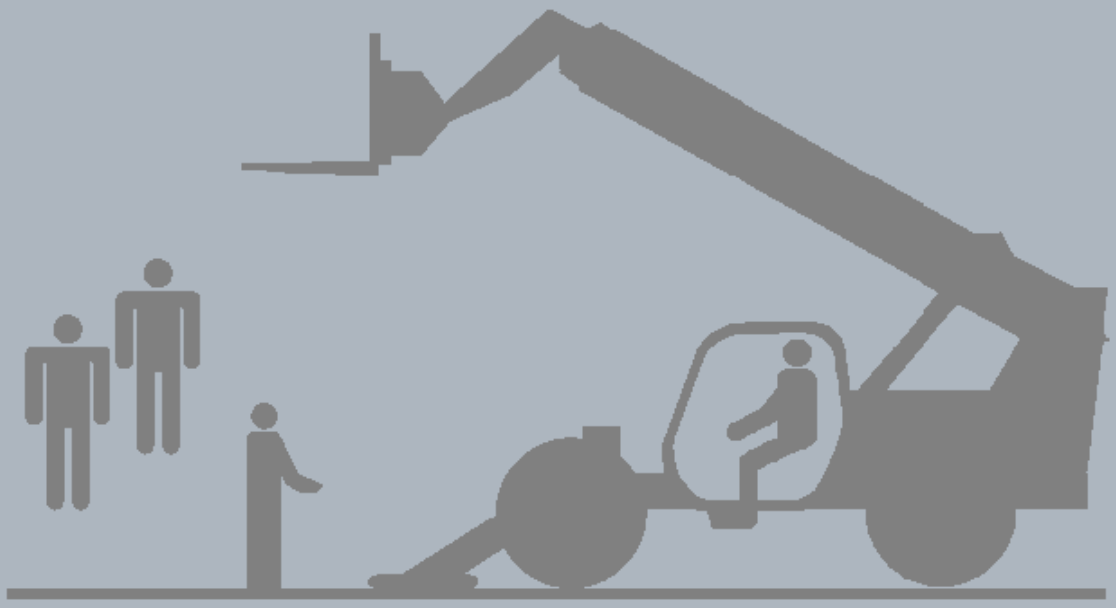
REGULATIONS & OPERATOR TRAINING



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Overview of Training Requirements

Summary

USA

ANSI/ITSDF

In the USA, the American National Standards Institute and the Industrial Truck Standards Development Foundation have established the safety requirements relating to the elements of design, operation, and maintenance of powered industrial trucks. The recognized standard in the USA for telehandlers is ANSI/ITSDF B56.6 and is free to download and print from the ITSDF web site at www.itsdf.org.

ANSI/ITSDF states “Only trained and authorized persons shall be permitted to operate a rough terrain forklift truck. Operators of rough terrain forklift trucks shall be qualified as to their visual, auditory, physical, and mental ability to operate the equipment safely. Furthermore, the standard prescribes certain responsibilities for the user’s and operators of the equipment pertaining to its safe operation and proper maintenance.

OSHA

The Occupational Safety and Health Administration, or OSHA, has clarified the training requirements for rough terrain forklift operator training. It was added to *Title 29 Code of Federal Regulations Part 1910.178.*

The clarification was issued to improve the training of operators with the goal of reducing the number of injuries and deaths that occur as a result of inadequate operator training.

These changes apply to both general industry and the construction industry.

• An _____of each powered industrial truck operator’s performance shall be conducted at least ***once every*** _____ ***years***.

• Refresher training must be conducted to ensure that the operator has the knowledge and skills needed to operate the powered industrial truck safely if one of the following conditions are true:

1. The operator has been observed to operate the vehicle in an unsafe manner;
2. The operator has been involved in an accident or near-miss incident;
3. The operator has received an evaluation that reveals that the operator is not operating the truck safely;
4. The operator is assigned to drive a different type of truck; or
5. A condition in the workplace changes in a manner that could affect safe operation of the truck.

Canada

CSA

Much like ANSI/ITSDF in the USA, the Canadian Standards Association, or CSA, has published Canada’s governing standard, CSA B335, which also establishes the safety requirements relating to the elements of design, operation, and maintenance of telehandlers.

CSA B335 standards do require operator retraining intervals not to exceed:

_____ years.

OHS

In Canada, CSA B335 standards for telehandlers are enforced by OHS, which is similar to OSHA in the USA, and other provincial authorities. Since each province has their own OHS, its important to be familiar with those rules and regulations that apply to the safe operation and maintenance of the telehandler.

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Requirement to Create a Safe Use Plan

Summary

The ANSI/ITSDF B56.6 Safety Standards require that a written safe use program specific to rough terrain forklift (RTFL) trucks shall be developed by the user.

Specifics

The Safe Use Plan must include, but not be limited to the following:

a) performing a site risk assessment to identify hazards, evaluate_____, develop control measures and communicate the safe use program to affected persons;

b) verifying that the RTFL truck and attachment(s) being utilized are appropriate for the application;

c) planning for access, preparation and maintenance of the site, as required, to include an assessment that the _____ _____is adequate to support the weight of the RTFL truck;

d) verifying that the RTFL truck _____, maintenance and repairs are performed per the provisions of this standard and as recommended by the manufacturer;

e) ensuring that only qualified and authorized personnel trained in accordance with OSHA 1910.178 (l) are allowed to operate the RTFL truck;

f) ensuring the _____ of trained, authorized RTFL truck operator(s) with the specific RTFL truck to be used;

g) informing the operator of local site requirements and warning and providing the means to protect against identified _____ in the areas where the RTFL truck will be operated;

h) monitoring the work being performed to ensure _____with provisions of this standard;

i) preventing unauthorized _____of the RTFL truck;

j) considering safety of persons in the vicinity of and _____ involved in the operation of the RTFL truck.

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OSHA 1910.178 (l)

Operator Training

• 1910.178(l)(1)

Safe Operation

• 1910.178(l)(1)**(i)**

The employer shall ensure that each powered industrial truck operator is competent to operate a powered industrial truck safely, as demonstrated by the successful completion of the training and evaluation specified in this paragraph (l).

• 1910.178(l)(1)**(ii)**

Prior to permitting an employee to operate a powered industrial truck (except for training purposes), the employer shall ensure that each operator has successfully completed the training required by this paragraph (l), except as permitted by paragraph (l)(5).

• 1910.178(l)(2)

Training Program Implementation

• 1910.178(l)(2)**(i)**

Trainees may operate a powered industrial truck only:

• 1910.178(l)(2)(i)**(A)**
Under the direct supervision of persons who have the knowledge, training, and experience to train operators and evaluate their competence; and

• 1910.178(l)(2)(i)**(B)**
Where such operation does not endanger the trainee or other employees.

• 1910.178(l)(2)**(ii)**
Training shall consist of a combination of formal instruction (lecture, discussion, interactive computer learning, video tape, written material), practical training (demonstrations performed by the trainer and practical exercises performed by the trainee), and evaluation of the operator's performance in the workplace.

• 1910.178(l)(2)(iii)

All operator training and evaluation shall be conducted by persons who have the knowledge, training, and experience to train powered industrial truck operators and evaluate their competence.

• 1910.178(l)(3)

Training Program Content

Powered industrial truck operators shall receive initial training in the following topics, except in topics which the employer can demonstrate are not applicable to safe operation of the truck in the employer's workplace.

• 1910.178(l)(3)**(i)**

Truck-Related Topics:

• 1910.178(l)(3)(i)**(A)**

Operating instructions, warnings, and precautions for the types of truck the operator will be authorized to operate;

• 1910.178(l)(3)(i)**(B)**

Differences between the truck and the automobile;

• 1910.178(l)(3)(i)**(C)**
Truck controls and instrumentation: where they are located, what they do, and how they work;

• 1910.178(l)(3)(i)**(D)**

Engine or motor operation;

• 1910.178(l)(3)(i)**(E)**

Steering and maneuvering;

• 1910.178(l)(3)(i)**(F)**

Visibility (including restrictions due to loading);

• 1910.178(l)(3)(i)**(G)**

Fork and attachment adaptation, operation, and use limitations;

• 1910.178(l)(3)(i)**(H)**

Vehicle capacity;

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OSHA 1910.178 (l)

• 1910.178(l)(3)(i)**(I)**

Vehicle stability;

• 1910.178(l)(3)(i)**(J)**

Any vehicle inspection and maintenance that the operator will be required to perform;

• 1910.178(l)(3)(i)**(K)**

Refueling and/or charging and boosting/jumping of battery;

• 1910.178(l)(3)(i)**(L)**

Operating limitations;

• 1910.178(l)(3)(i)**(M)**

Any other operating instructions, warnings, or precautions listed in the operator's manual for the types of vehicle that the employee is being trained to operate.

• 1910.178(l)(3)**(ii)**

Workplace-Related Topics

• 1910.178(l)(3)(ii)**(A)**

Surface conditions where the vehicle will be operated;

• 1910.178(l)(3)(ii)**(B)**

Composition of loads to be carried and load stability;

• 1910.178(l)(3)(ii)**(C)**

Load manipulation, stacking, and unstacking;

• 1910.178(l)(3)(ii)**(D)**

Pedestrian traffic in areas where the vehicle will be operated;

• 1910.178(l)(3)(ii)**(E)**

Narrow aisles and other restricted places where the vehicle will be operated;

• 1910.178(l)(3)(ii)**(F)**

Hazardous (classified) locations where the vehicle will be operated;

• 1910.178(l)(3)(ii)**(G)**

Ramps and other sloped surfaces that could affect the vehicle's stability;

• 1910.178(l)(3)(ii)**(H)**

Closed environments and other areas where insufficient ventilation or poor vehicle maintenance could cause a buildup of carbon monoxide or diesel exhaust;

• 1910.178(l)(3)(ii)**(I)**

Other unique or potentially hazardous environmental conditions in the workplace that could affect safe operation.

• 1910.178(l)(3)**(iii)**

The requirements of this Section.

• 1910.178(l)(4)

Refresher Training and Evaluation

• 1910.178(l)(4)**(i)**

Refresher training, including an evaluation of the effectiveness of that training, shall be conducted as required by paragraph (l)(4)(ii) to ensure that the operator has the knowledge and skills needed to operate the powered industrial truck safely.

• 1910.178(l)(4)**(ii)**

Refresher training in relevant topics shall be provided to the operator when:

• 1910.178(l)(4)(ii)**(A)**

The operator has been observed to operate the vehicle in an unsafe manner;

• 1910.178(l)(4)(ii)**(B)**

The operator has been involved in an accident or near-miss incident;

• 1910.178(l)(4)(ii)**(C)**

The operator has received an evaluation that reveals that the operator is not operating the truck safely;

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OSHA 1910.178 (l)

• 1910.178(l)(4)(ii)**(D)**

The operator is assigned to drive a different type of truck; or

• 1910.178(l)(4)(ii)**(E)**

A condition in the workplace changes in a manner that could affect safe operation of the truck.

• 1910.178(l)(4)**(iii)**

An evaluation of each powered industrial truck operator's performance shall be conducted at least once every three years.

• 1910.178(l)(5)

Avoidance of Duplicate Training

If an operator has previously received training in a topic specified in paragraph (l)(3) of this section, and such training is appropriate to the truck and working conditions encountered, additional training in that topic is not required if the operator has been evaluated and found competent to operate the truck safely.

• 1910.178(l)(6)

Certification

The employer shall certify that each operator has been trained and evaluated as required by this paragraph (l). The certification shall include the name of the operator, the date of the training, the date of the evaluation, and the identity of the person(s) performing the training or evaluation.

• 1910.178(l)(7)

Dates

The employer shall ensure that operators of powered industrial trucks are trained, as appropriate, by the dates shown in the following table.

If the employee was hired:

Before December 1, 1999
The initial training and evaluation of that employee must be completed:

By December 1,1999

If the employee was hired:

After December 1, 1999

The initial training and evaluation of that employee must be completed:

Before the operator is assigned to operate a powered industrial truck.

• 1910.178(l)(8)

Appendix A to this section provides non-mandatory guidance to assist employers in implementing this paragraph (l). This appendix does not add to, alter, or reduce the requirements of this section.

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OSHA 1910.178 (l)

Appendix A — Stability of Powered Industrial Trucks (Non-mandatory Appendix to Paragraph (l) of This Section)

A-1. **Definitions.**

The following definitions help to explain the principle of stability:

Center of gravity is the point on an object at which all of the object's weight is concentrated. For symmetrical loads, the center of gravity is at the middle of the load.

Counterweight is the weight that is built into the truck's basic structure and is used to offset the load's weight and to maximize the vehicle's resistance to tipping over.

Fulcrum is the truck's axis of rotation when it tips over.

Grade is the slope of a surface, which is usually measured as the number of feet of rise or fall over a hundred foot horizontal distance (the slope is expressed as a percent).

Lateral stability is a truck's resistance to overturning sideways.

Line of action is an imaginary vertical line through an object's center of gravity.

Load center is the horizontal distance from the load's edge (or the fork's or other attachment's vertical face) to the line of action through the load's center of gravity.

Longitudinal stability is the truck's resistance to overturning forward or rearward.

Moment is the product of the object's weight times the distance from a fixed point (usually the fulcrum). In the case of a powered industrial truck, the distance is measured from the point at which the truck will tip over to the object's line of action. The distance is always measured perpendicular to the line of action.

Track is the distance between the wheels on the same axle of the truck.

Wheelbase is the distance between the centerline of the vehicle's front and rear wheels.

A-2. **General.**

A-2.1. Determining the stability of a powered industrial truck is simple once a few basic principles are understood. There are many factors that contribute to a vehicle's stability: the vehicle's wheelbase, track, and height; the load's weight distribution; and the vehicle's counterweight location (if the vehicle is so equipped).

A-2.2. The "stability triangle," used in most stability discussions, demonstrates stability simply.

A-3. **Basic Principles.**

A-3.1. Whether an object is stable depends on the object's moment at one end of a system being greater than, equal to, or smaller than the object's moment at the system's other end. This principle can be seen in the way a see-saw or teeter-totter works: that is, if the product of the load and distance from the fulcrum (moment) is equal to the moment at the device's other end, the device is balanced and it will not move. However, if there is a greater moment at one end of the device, the device will try to move downward at the end with the greater moment.

A-3.2. The longitudinal stability of a counterbalanced powered industrial truck depends on the vehicle's moment and the load's moment. In other words, if the mathematic product of the load moment (the distance from the front wheels, the approximate point at which the vehicle would tip forward) to the load's center of gravity times the load's weight is less than the vehicle's moment, the system is balanced and will not tip forward. However, if the load's moment is greater than the vehicle's moment, the greater load-moment will force the truck to tip forward.

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OSHA 1910.178 (l)

A-4. **The Stability Triangle.**

A-4.1. Almost all counterbalanced powered industrial trucks have a three-point suspension system, that is, the vehicle is supported at three points. This is true even if the vehicle has four wheels. The truck's steer axle is attached to the truck by a pivot pin in the axle's center. When the points are connected with imaginary lines, this three-point support forms a triangle called the stability triangle.

A-4.2. When the vehicle's line of action, or load center, falls within the stability triangle, the vehicle is stable and will not tip over. However, when the vehicle's line of action or the vehicle/ load combination falls outside the stability triangle, the vehicle is unstable and may tip over.

A-5. **Longitudinal Stability.**

A-5.1. The axis of rotation when a truck tips forward is the front wheels' points of contact with the pavement. When a powered industrial truck tips forward, the truck will rotate about this line. When a truck is stable, the vehicle-moment must exceed the load-moment. As long as the vehicle-moment is equal to or exceeds the load-moment, the vehicle will not tip over. On the other hand, if the load moment slightly exceeds the vehicle-moment, the truck will begin to tip forward, thereby causing the rear to lose contact with the floor or ground and resulting in loss of steering control. If the load-moment greatly exceeds the vehicle moment, the truck will tip forward.

A-5.2. To determine the maximum safe load-moment, the truck manufacturer normally rates the truck at a maximum load at a given distance from the front face of the forks. The specified distance from the front face of the forks to the line of action of the load is commonly called the load center. Because larger trucks normally handle loads that are physically larger, these vehicles have greater load centers. Trucks with a capacity of 30,000 pounds or less are normally rated at a given load weight at a 24-inch load center. Trucks with a capacity greater than 30,000 pounds are normally rated at a given load weight at a 36- or 48-inch load center. To safely operate the vehicle, the operator should always check the data plate to determine the maximum allowable weight at the rated load center.

A-5.3. Although the true load-moment distance is measured from the front wheels, this distance is greater than the distance from the front face of the forks. Calculating the maximum allowable load-moment using the load-center distance always provides a lower load- moment than the truck was designed to handle. When handling unusual loads, such as those that are larger than 48 inches long (the center of gravity is greater than 24 inches) or that have an offset center of gravity, etc., a maximum allowable load-moment should be calculated and used to determine whether a load can be safely handled. For example, if an operator is operating a 3000 pound capacity truck (with a 24- inch load center), the maximum allowable load-moment is 72,000 inch-pounds (3,000 times 24). If a load is 60 inches long (30-inch load center), then the maximum that this load can weigh is 2,400 pounds (72,000 divided by 30).

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OSHA 1910.178 (l)

A-6. **Lateral Stability.**

A-6.1. The vehicle's lateral stability is determined by the line of action's position (a vertical line that passes through the combined vehicle's and load's center of gravity) relative to the stability triangle. When the vehicle is not loaded, the truck's center of gravity location is the only factor to be considered in determining the truck's stability. As long as the line of action of the combined vehicle's and load's center of gravity falls within the stability triangle, the truck is stable and will not tip over. However, if the line of action falls outside the stability triangle, the truck is not stable and may tip over.

A-6.2. Factors that affect the vehicle's lateral stability include the load's placement on the truck, the height of the load above the surface on which the vehicle is operating, and the vehicle's degree of lean.

A-7. **Dynamic Stability.**

A-7.1. Up to this point, the stability of a powered industrial truck has been discussed without considering the dynamic forces that result when the vehicle and load are put into motion. The weight's transfer and the resultant shift in the center of gravity due to the dynamic forces created when the machine is moving, braking, cornering, lifting, tilting, and lowering loads, etc., are important stability considerations.

A-7.2. When determining whether a load can be safely handled, the operator should exercise extra caution when handling loads that cause the vehicle to approach its maximum design characteristics. For example, if an operator must handle a maximum load, the load should be carried at the lowest position possible, the truck should be accelerated slowly and evenly, and the forks should be tilted forward cautiously. However, no precise rules can be formulated to cover all of these eventualities.

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CSA B335

CSA B335 Clause 6 (partial) –

There are a number of sub-sections within Part 6 of the CSA B335 standard. We'll only be covering a few regarding training requirements. As always its important to be familiar with all of the requirements as it relates to the safe operation and maintenance of the telehandler.

Operator Training Requirements

6.1 General

Clause 6 specifies the knowledge, practical, and evaluation requirements of a lift truck training program. Subjects in this Clause specify content requirements as well as a suggested method of instruction and evaluation.

The trainee shall be instructed on the safe operational procedures of a lift truck in accordance with the manufacturer's operator manual, the user's operating guidelines, and government regulations. Instruction shall be given on the following:

(a) legislation;

(b) company policies and procedures;

(c) operating manual;

(d) lift truck features and safety equipment;

(e) stability;

(f) capacity plate and location;

(g) preoperational inspection;

(h) start-up;

(i) travelling, with and without a load;

(j) pedestrians;

(k) load handling;

(l) ramps and grades;

(m) personnel lifting, lowering, and supporting;

(n) elevators;

(o) workplace specific hazards;

(p) procedures for shutdown/leaving the operator's position; and

(q) refueling/recharging.

6.20 Evaluations

6.20.1 General

The trainee shall be evaluated throughout the learning process to determine whether the objectives of the training course are being met.

6.21.1 Retraining

Operators of lift trucks shall receive retraining at intervals not exceeding three years. Retraining shall meet the requirements of Clause 6.

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CSA B335

6.21.3 Upgrade training

Upgrade training shall be given to lift truck operators when

(a) equipment is introduced in the workplace that is unfamiliar to the operator;

(b) the equipment used by the operator is modified;

(c) the operating conditions or the environment in which the operator works is changed (e.g., the operator works in a different area, moves different types of loads, etc.);

(d) applicable legislation changes; or

(e) skill or knowledge deficiencies have been identified. The upgrade training course shall include all relevant information, as applicable. The content, delivery method, and individual learning needs of the operator shall determine the course length. Knowledge verification and practical evaluation shall be performed in accordance with Clause 6.20.

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FIVE PRINCIPLES OF SAFE OPERATION

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Introduction

It is the responsibility of the _____ to learn and practice the principles of safe machine operation contained in this training manual and the machine's operator's manual. The five principles of safe operation are:

1. Avoid hazardous situations

2. Always perform a pre-operation inspection

3. Always perform function tests prior to use

4. Inspect the workplace

5. Only use the machine as it was intended

Each principle of safe operation will be discussed in detail in the following sections.

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1. Avoid Hazardous Situations

Safety Signs

Safety signs are used to identify potential hazards. Safety signs are placed on the equipment in strategic locations so that the operator has easy access to safety information.

All safety signs are intended to convey important safety information to the personnel working on and around the machine. It is the _____ responsibility to read, understand and obey the information contained in these safety signs for their safety as well as those working in the area.

Hazard Classification

Genie uses symbols, color coding and signal words to identify the following:

Safety alert symbol is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

Red

Indicates a hazardous situation that, if not avoided, will result in death or serious injury.

Orange

Indicates a hazardous situation that, if not avoided, could result in death or serious injury.

Yellow

Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury.

Blue

Indicates information considered important, but not hazard related
(e.g., messages related to property damage)

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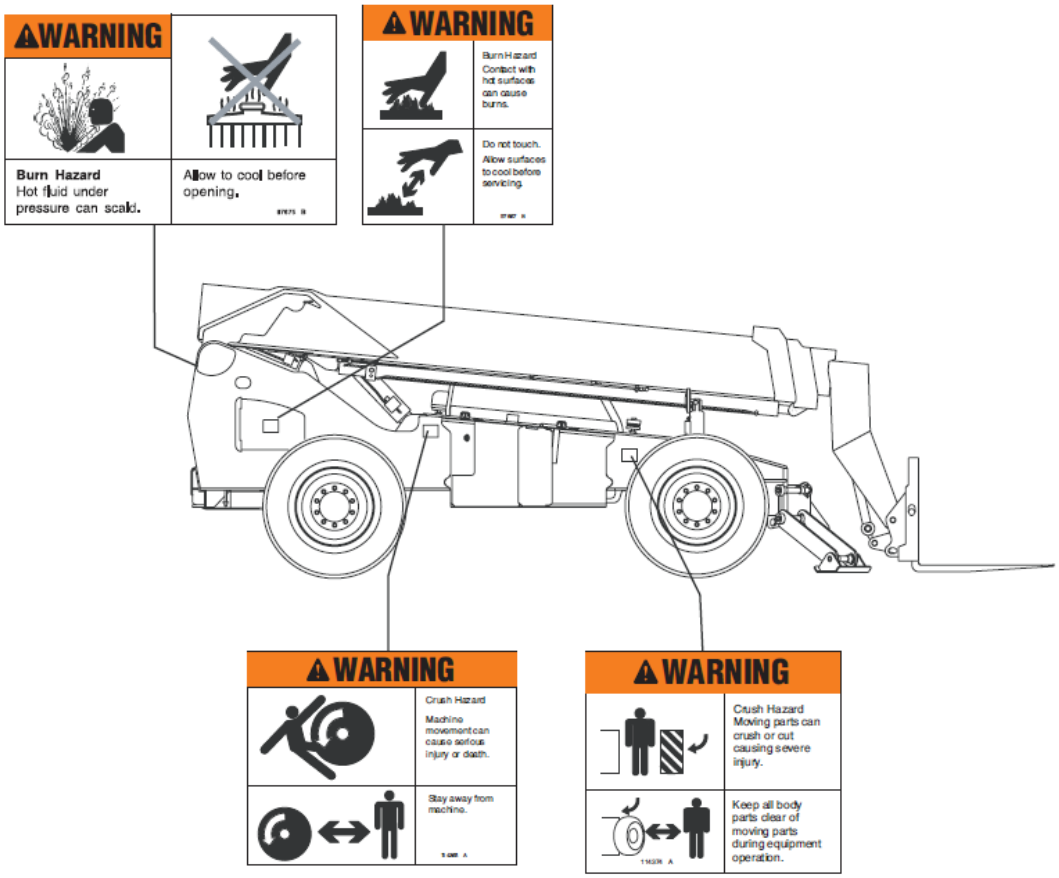
Examples

The following are examples from a telehandler. Notice the placement of the decals in the areas where the potential hazard exists.

This is only an example. Specific hazard information can be found in the operator's manual for the machine you will be using.



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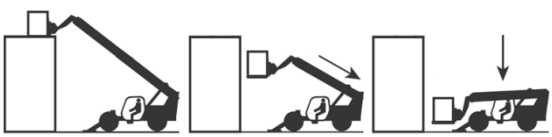
Work Area Safety

Telehandler operators should be familiar with the following operational considerations before operating the machine.

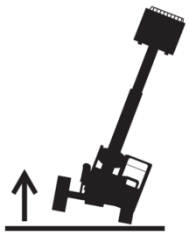
For more details on potential hazardous situations associated with the operation of a telehandler, refer to the appropriate operator's manual provided by the manufacturer.

Overturning Hazards

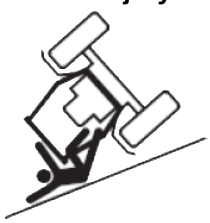
- Always wear a _____ _____when operating the machine.
- Use the _____ _____to confirm that the load is within the rated capacity of the machine. Do not exceed the rated load.
- Do not operate the machine if the load chart is missing.
- Do not raise a load and then drive to position it.
- Do not lower a load without _____ the boom first.



- Do not operate the machine while using front-end attachments if the capacity ratings or other critical decals or markings are missing.
- Do not raise the load unless the ground can support all forces imposed by the machine and the load.
- Do not raise the boom unless the machine is _____. The machine level indicator should be at _____ degrees.
- Do not raise a load unless the load is properly positioned or secured on the forks or approved attachment
- Do not operate the machine in strong or gusty winds. Do not increase the surface area of the carriage or the load. Increasing the surface area exposed to the wind will _____machine stability.



- Do not raise the boom unless the parking brake is set.
- Do not level the machine using the frame sway controls unless the boom angle indicator is at zero degrees.
- Do not use the sway control to position an elevated load.
- An over inflated tire can explode which may compromise machine stability and cause the machine to tip over.
- The use of temporary flat tire repair products may also lead to tire failure which could compromise machine stability and cause the machine to tip over.
- Use extreme care and slow speeds while driving the machine in the travel position across uneven terrain, debris, unstable or slippery surfaces and near holes and drop-offs.
- **Should the machine tip over, grasp the steering wheel tightly and lean away from the point of impact. Never attempt to jump clear of the machine. Doing so could result in death or serious injury.**



Collision Hazards

- Do not put the transmission into gear unless the parking brake is set or the service brake is applied.
- Do not drive the machine if visibility is _____.
- Do not operate the machine with a faulty backup alarm.
- Operate the machine at speeds that will keep the load under control. Start and stop movements smoothly.
- Maintain a safe distance from edges of ramps or elevated platforms.

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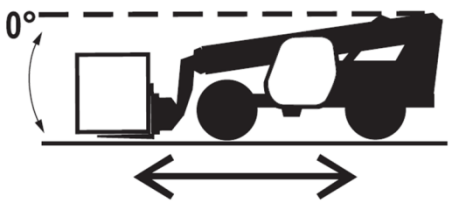
- Set the parking brake, put the transmission in neutral and lower the carriage or the attachment to the ground before leaving the machine.
- Do not _____ telehandlers or other construction equipment at intersections or in areas where the view may be obstructed.
- When driving, keep the boom at or below horizontal and keep the load close to the ground.



Component Damage

- Do not use a damaged or malfunctioning machine.
- Do not use any battery or charger greater than 12V to jump-start the engine.
- Do not use the machine as a ground for welding.
- Do not drive the machine unless the stabilizers are fully retracted.
- Do not alter or disable machine components that in any way affect safety or stability.
- Do not attempt to start the machine by towing or pushing.
- Be sure all decals are in place and legible.

Health and Safety

- When getting in and out of the cab, face the machine, use the steps and handrails provided and always maintain _____-_____ contact.
- Do not use the steering wheel or any other controls as handrails.
- Always adjust the seat and fasten the seat belt before starting the engine.
- Always wear a seat belt when operating the machine.
- Always remain completely _____ the cab when operating the machine.
- Operators must comply with employer, job site and governmental rules regarding the use of personal protective equipment.
- Overhead guards may be offered on telehandlers for protecting the operator from falling objects. Overhead guards can only protect against small objects and are not to be considered as a substitute to good load management habits.

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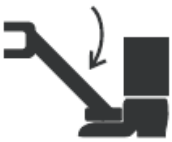
Crush Hazards

- Keep clear of moving parts during machine operation.
- Keep clear of moving stabilizers.



- Keep clear of elevated components.
- Never insert any part of your _____ into moving machine parts, such as the boom structure.
- Ensure that all ground personnel stay clear of the machine during operation.

Explosion and Fire Hazards

- Do not start the engine if you smell or detect liquid petroleum gas (LPG), gasoline, diesel fuel or other explosive substances.
- Do not refuel the machine with the engine running. Keep all open flames and sparks away.
- Refuel the machine and charge the battery only in an open, well-ventilated area away from sparks, flames and lighted tobacco.

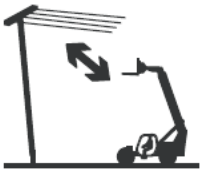
- Do not operate the machine in hazardous locations or locations where potentially flammable or explosive gases or particles may be present.
- Do not spray ether into engines equipped with glow plugs or air intake grid heaters.

Burn Hazards

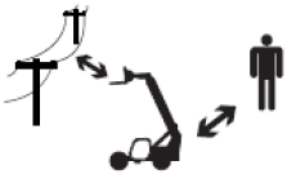
- Allow hot surfaces to cool before touching or servicing.

- Improper contact with components under any cover will cause serious injury. Only trained maintenance personnel should access compartments. Access by the operator is only advised when performing a pre-operation inspection. All compartments must remain closed and secured during operation.

Electrocution Hazards



- Telehandlers are not electrically insulated and provide _____ protection from electrocution.
- Keep away from the telehandler if it contacts energized power lines.



- Working near energized power sources requires common sense and planning. To help with the planning, there is information in the operator's manual and on_____ attached to the telehandler that will identify, at a minimum, the required clearance you must maintain when working near electrical conductors. Do not operate a machine without this information.

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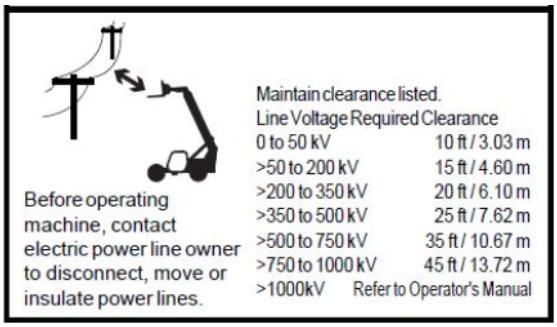
- It is mandatory that all operators fully understand the potential for electrocution when operating a telehandler. High voltage overhead power lines are not insulated. The operator or any part of the machine will not have to come in direct contact with a power line to be in extreme danger.

- High voltage electrical current can jump up to _____feet to seek a ground.

Operators must know, or find out, the voltage of all power sources in their work area and maintain the _____ _____in accordance with the information in the operator's manual and on decals attached to the telehandler.



Example of required clearance information

- Always contact the electric power line owner. The electric power must be disconnected or the power lines moved or insulated before machine operations begin.

NOTE : Refer to the Manufacturers Operators Manual for more information on potential hazards and operational considerations.

Maintenance and Repair

- Be sure the operator's and safety manuals are complete, legible and in the storage container located _____the cab.

- Be sure all decals are in place and legible.

- Do not replace items critical to machine stability with items of different weight or specification.

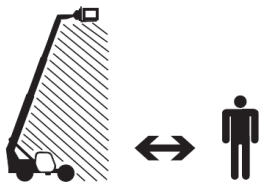
- Do not replace factory-installed tires with tires of different specification or ply rating.

- Do not use a damaged or malfunctioning machine.

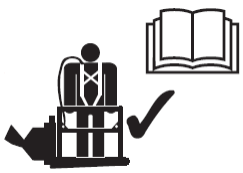
- Be sure all maintenance has been performed as specified in the appropriate manufacturer's service manual.

Personnel Safety

- Keep people, equipment and material out of the work area. Do not operate the machine while people are under or near an elevated boom, whether it is loaded or unloaded.



- Do not transport or lift personnel with a telehandler unless it is equipped with a manufacturer approved work platform.



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Additional Safety Considerations

- A load backrest extension should be used as necessary to prevent loads from falling toward the operator.

- Do not operate the machine with a hydraulic oil or air leak. Escaping fluid under pressure can penetrate skin, causing serious injury. Relieve pressure before disconnecting hydraulic lines. Keep away from leaks and pin holes. Use a piece of cardboard or paper to search for leaks.

  Do not use your hand.

- Fluid injected into skin must be surgically removed within a few hours by a doctor familiar with this type of injury or gangrene will result.

- Always operate the machine in a well-ventilated area to avoid carbon monoxide poisoning.

- Use proper hearing protection if the work site or the inside of the telehandler cab has high noise levels.

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2. Always Perform a Pre-Operation Inspection

Fundamentals

It is the responsibility of the _____ to perform a pre-operation inspection and, if required by the employer, routine maintenance. The operator should wear the required personal protective equipment during the inspection as required by jobsite rules and regulations.

The pre-operation inspection is a visual inspection performed by the operator _____ to each work shift. The inspection is designed to discover if anything is apparently wrong with a machine before the operator performs the function tests.

The pre-operation inspection also serves to determine if routine maintenance procedures are required. Only routine maintenance items specified in the Operator's Manual may be performed by the operator (and only when they are authorized by the employer to do so).

If damage or any unauthorized variation from factory delivered condition is discovered, the machine must be _____ and _____from service.

Repairs to the machine may only be made by a qualified service technician, according to the manufacturer's specifications. After repairs are completed, the operator must perform a pre-operation inspection again before going on to the function tests.

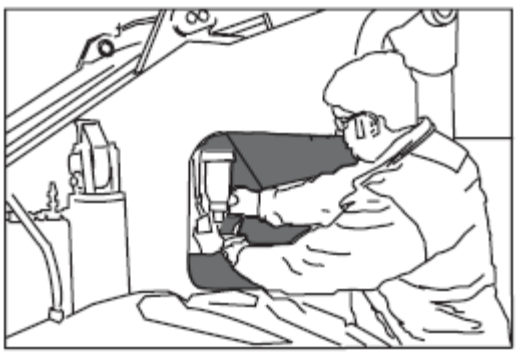
Scheduled maintenance inspections must be performed by qualified service technicians, according to the manufacturer's specifications.

Pre-Operation Inspection Examples

Some examples of items you need to inspect as part of the pre-operation inspection are:

- Check that the Operator's Manual is in the storage compartment inside the cab
- Check around and under the machine for oil, hydraulic or coolant leaks
- Check oil level, coolant level, and if trained and authorized, battery level. Always wear the proper personal protective equipment when checking batteries.
- Check tire air pressure
- Check for loose or missing parts and fasteners, unauthorized modifications, and damaged or improperly installed components
- Inspect the general condition of the machine. Look for cracks, dents, excessive rust or corrosion, and other structural damage.
- Check for missing decals.

Please refer to the detailed pre- operational checklist provided in the Operator's Manual for thorough inspection of the machine's condition.



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3. Always Perform Function Tests Prior To Use

Fundamentals

The function tests are designed to discover any malfunctions _____ the machine is put into service. The operator must follow the step-by-step instructions to test all machine functions.

A malfunctioning machine must _____ be used. If malfunctions are discovered, the machine must be tagged and removed from service.

Repairs to the machine may only be made by a qualified service technician, according to the manufacturer's specifications.

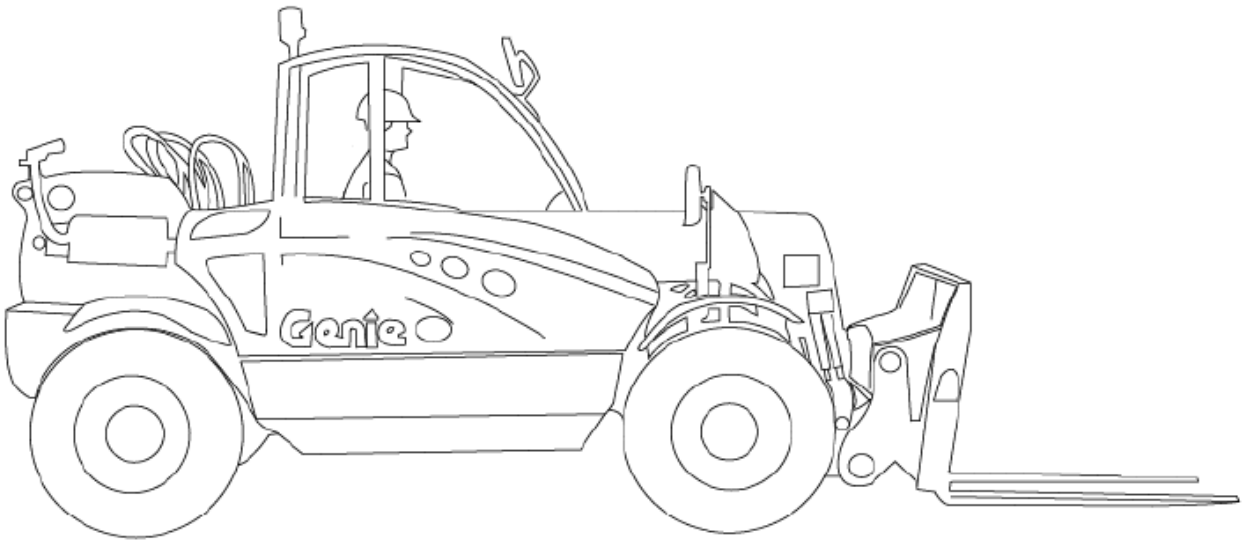
After repairs are completed, the operator must perform a pre-operation inspection and function tests again before putting the machine into service.

Function Test Examples

Some examples of functions tests that you need to perform are:

- Test all controls
- Test the safety features
- Test the steering modes
- Test the parking and service brakes
- Test the stabilizers
- Test the attachment

Please refer to the detailed function test checklist provided in the Operator's Manual for thorough inspection of the machine's condition.



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4. Inspect The Workplace

Fundamentals

The workplace inspection helps the operator determine if the workplace is suitable for safe machine operation. It should be performed by the operator _____ to moving the machine to the workplace.

The operator should always wear the proper personal protective equipment as required by jobsite rules or the jobsite conditions or hazards that justify it.

It is the operator's responsibility to _____ and _____workplace hazards, then watch for and avoid them while moving, setting up and operating the machine.

Pre-Operation Inspection Examples

Be aware of and avoid the following hazardous situations:

- Drop-offs or holes
- Bumps, floor obstructions or debris
- Sloped surfaces
- Unstable or slippery surfaces
- Overhead obstructions
- High voltage conductors
- Specific hazards such as flammable liquids, combustible dusts and explosive atmospheres
- Inadequate surface support to withstand all forces imposed by the machine and the load
- Wind and bad weather conditions
- The presence of unauthorized personnel
- Accumulation of exhaust emissions in the workplace, operator's cab or other restricted spaces
- High sound level in the work area
- All other possibly unsafe conditions

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5. Only Use The Machine As It Was Intended

The Operating Instructions section provides instructions for each aspect of telehandler operation.

It is the operator's responsibility to follow all the safety rules and instructions in the Operator's and Safety manuals.

Only _____and _____personnel should be permitted to operate a telehandler.

If more than one operator is expected to use a telehandler at different times in the same work shift, they must all be qualified operators and are expected to follow all safety rules and instructions in the Operator's and Safety manuals.

That means _____ _____ _____ should perform a pre-operation inspection, function tests, and a workplace inspection before using the machine.

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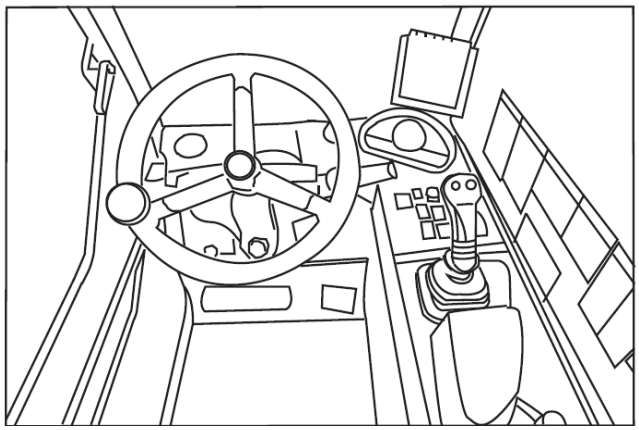
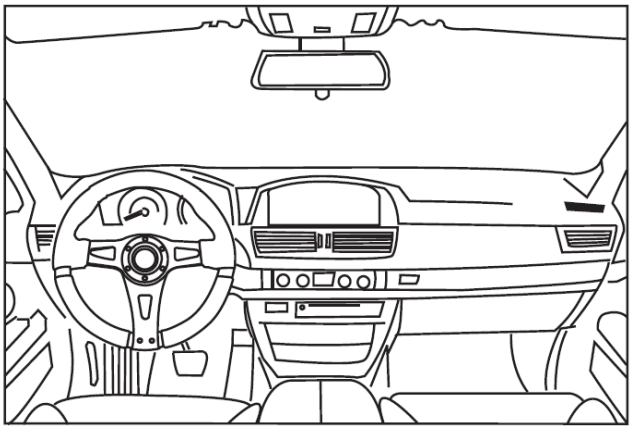
Fundamentals

Telehandler vs. Automobile



Telehandler



Automobile

Operators must keep in mind the similarities and, more importantly, the differences between automobiles and telehandlers.

Similarities

Both, automobiles and telehandlers:

- are equipped with four tires
- are engine powered
- have safety features
- have visibility obstructions
- utilize steering, brakes, horns, mirrors and seat belts

Besides the similarities mentioned above, automobiles and telehandlers _____ and _____very differently. The operator must be aware and alert at all times during machine operation.

Differences

Major differences which directly affect safety, machine operation and stability are:

Center of gravity: The center of gravity for telehandlers may change during operation, depending on the position of the machine, boom and the load. This can significantly affect the stability of the telehandler during operation. For automobiles, center of gravity is almost always constant.

Suspension: Telehandlers typically do not have any suspension systems. Automobiles have suspension systems designed to provide a comfortable ride.

Weight: Since telehandlers use large steel components as counterbalance, the weight of the telehandler is significantly greater than that of an average automobile.

Steering modes: Typically, only the front two wheels in automobiles assist in steering, whereas telehandlers have multiple steering modes (front wheel, four-wheel, crab steer and sometimes rear wheel).

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Stability: Telehandlers are unstable if the combined center of gravity falls outside the stability triangle. Automobiles have four points of stability, with all four wheels in contact with the ground.

Passengers: Passengers may or may not be allowed on a telehandler. Check with the machine manufacturer for more information.

Layout of controls: Telehandler machine controls are typically hydraulically piloted proportional controls. There may also be a number of different switches used to control functions such as chassis sway, work lights, auxiliary controls.

Road surface: Telehandlers are used primarily in _____ _____ environments.

Visibility: Telehandlers may have structures around the machine which could result in obstructions to visibility.

Speed: Telehandlers usually never exceed 20-25 miles an hour.

Stopping distance: A telehandler is typically much heavier than an automobile and therefore the stopping distance may be different at the same speed.

Transmission: Telehandlers will typically be driven either by a powershift or hydrostatic drive transmission.

Parking brake: Telehandlers will typically utilize a spring-applied, hydraulically-released parking brake system.

Visibility Considerations

Some conditions of reduced visibility when operating a telehandler are:

- when entering or exiting narrow passage ways, storage rack aisles
- other vehicles or the load itself

Whenever visibility is obstructed, drive slowly and use your horn to warn others.

Use additional personnel to guide you if you are unable to see over the load

Operators must be aware of all visibility restrictions _____ _____ and while operating the telehandler.

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Capacity Plates

The machine capacity rating, model name, model year, serial number and other machine related designations are captured on plates found on the telehandler.

These manufacturer's plates are typically located in _____ and easily accessible areas on the machine. Be aware that the location of the plates may vary, depending on the manufacturer and model type.

Machine or Attachment Modifications

A modified telehandler or attachment could potentially affect the machines overall capacity, stability or safe operation. That's why its extremely important to _____ _____ or alter a telehandler or its attachment without the telehandler manufacturers prior written approval.

Even using an attachment that is unapproved by the telehandler manufacturer is considered a modification of the machine. Always contact the telehandler manufacturer before using any attachment that has not been previously approved for use on the machine.

Example of a Manufacturer's Plate

Example of an Attachment Plate

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Load Charts

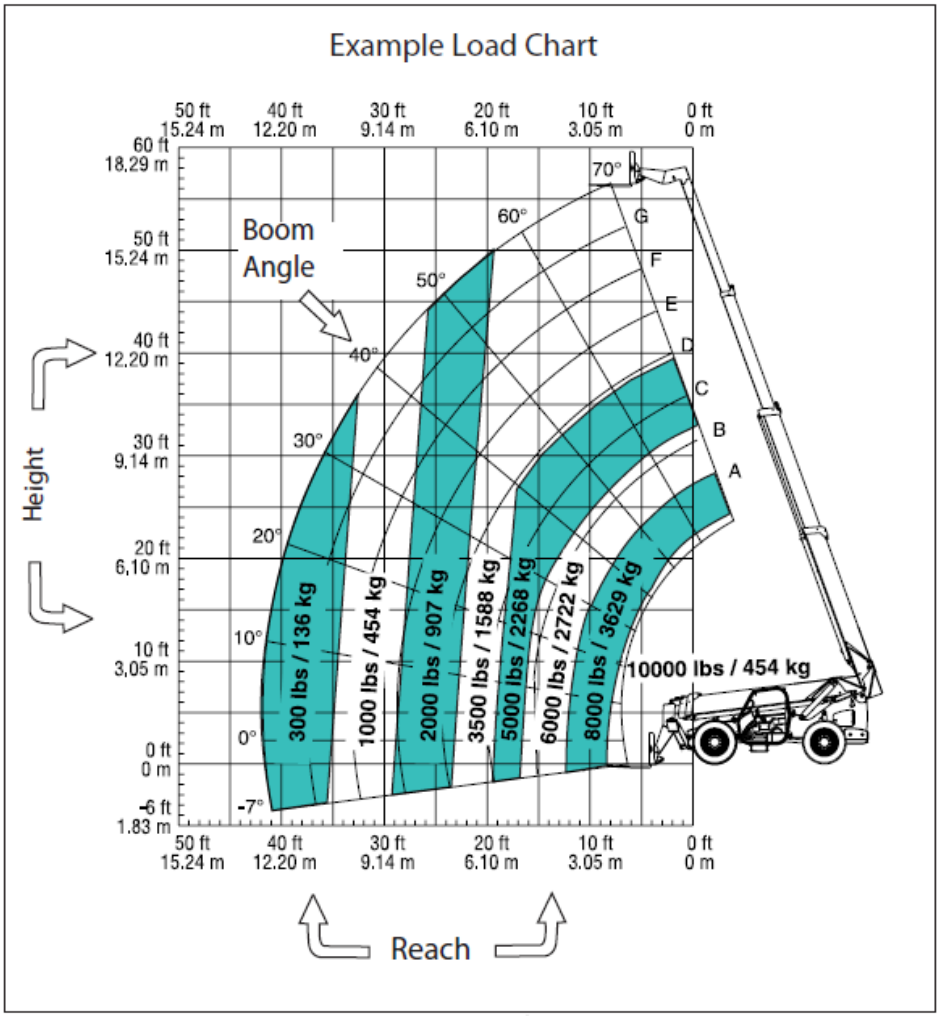
Load charts show the maximum _____ _____ at the various lifting heights and boom elevation angles.

The maximum allowable attachment capacity referenced on the load chart changes as the boom is raised and extended to maintain the stability of the telehandler.

Load charts are typically located in direct line of sight for the operator, inside the operator's cabin.

Load charts are based on a leveled telehandler that is on firm ground, capable of supporting the combined load of the telehandler and the object being lifted. Load charts may also provide additional information to help the operator determine the correct capacity for the attachment being used such as:

- if stabilizers are required
- if axle lock required
- Specific attachment model and serial number information
- Restricted attachment configuration or use



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Reading a Load Chart

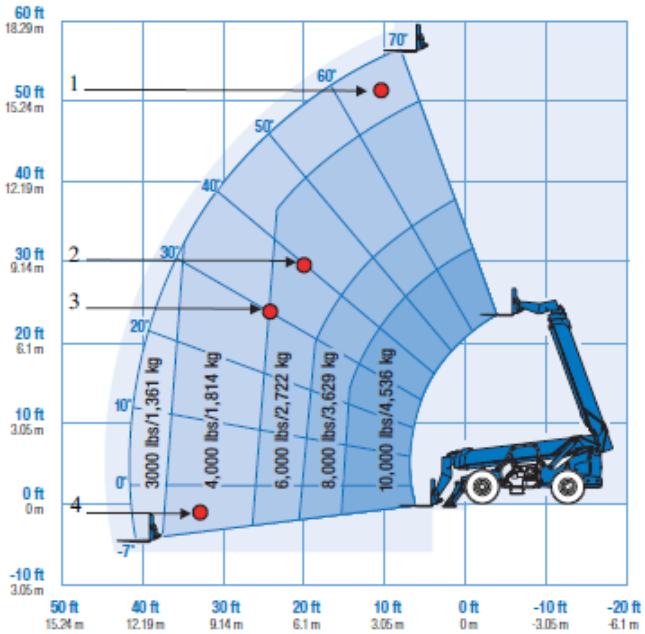
Load capacity varies depending on boom angle, boom extension and if stabilizers have been deployed, as well as the specific attachment being used.

Be sure that the load chart in the telehandler _____ the machine and the approved attachment on the machine before lifting the load.

Load charts should always be clearly legible and _____to the operator in the normal operating position.

Below are examples of four different lifting scenarios. See if you can determine if the machine is stable or unstable in each scenario.

Case	Load Wt	Horizontal Extension	Vertical Extension	Stability Condition
1	4000 lbs	10 ft (3 m)	50 ft (15.25 m)	
2	8000 lbs	20 ft (6 m)	28 ft (8.5 m)	
3	9000 lbs	26 ft (8 m)	23 ft (7 m)	
4	3000 lbs	33 ft (10 m)	-2 ft (-0.6 m)	



Load Selection

Always follow the machine's Operator's Manual for load selection information.

- Determine the weight of the load and ensure that it does not exceed the telehandler's or attachment's maximum capacity.
- Evaluate the dimensions of the load to determine proper load handling techniques.
- Avoid moving loads that may come apart. Always secure the load to minimize the potential for falling debris.
- Pallets and containers should be in good condition to ensure adequate support of the load.

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Load Handling

Stability

There are many conditions that may affect the stability of the machine during operation. Ground conditions or operating surface, surface grade, travel speed, wind conditions, weight or size of the load, tire air pressure, and operator judgment can all have an influence on stability, load handling and machine operation.

Fundamentals

Stability Triangle Definition

Three points on the machine, the two front tires (or stabilizers) and the oscillating pivot point at the center of the rear axle, form a triangle which is known as the *Stability Triangle* (see figure on right).

Center of Gravity Definition

Center of gravity, or C.G., is the single point on an object at which all of the object's weight is_____.

1. The telehandler has a center of gravity.

2. The load has a center of gravity.

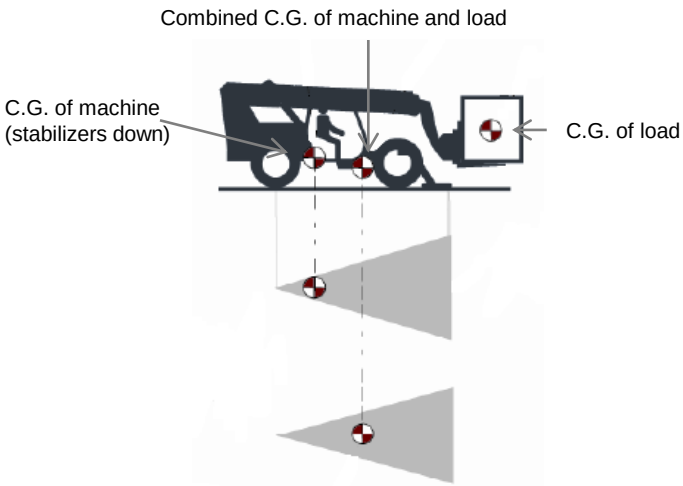
3. When a load is placed on the telehandler, there is a _____ center of gravity. If the center of gravity moves outside the stability triangle at any time during operation, the machine will become unstable and will tip over.

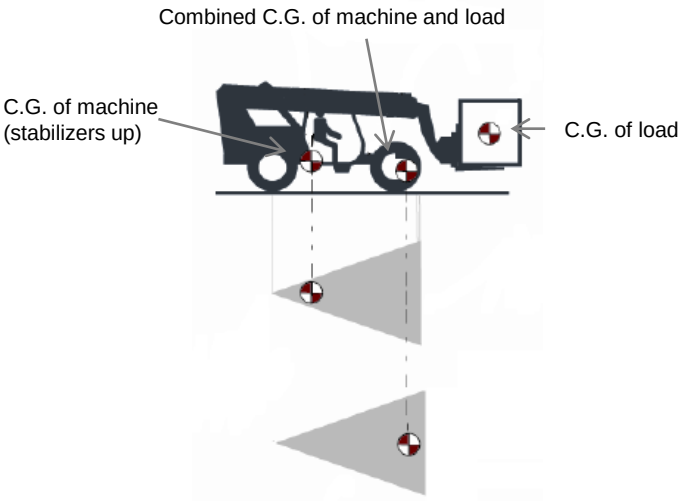
Stability Considerations During Operation

It is important to understand the theory of center of gravity during machine operation in order to avoid conditions of instability.

During machine operation, the location of the center of gravity can be dynamic. That is, the location can move under different operating conditions such as when traveling, braking, and lifting the load.

NOTE: Unless specifically mentioned, it is assumed that the operating surface is always a firm and level ground.





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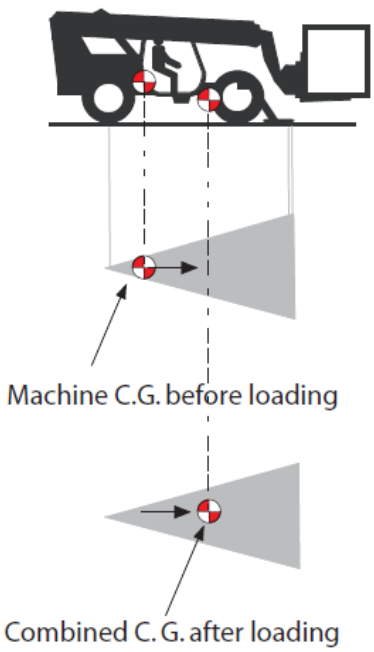
Loading in stowed position

When a telehandler is loaded in the stowed position, the center of gravity moves forward and lower, towards the front of the machine and closer to the operating surface.



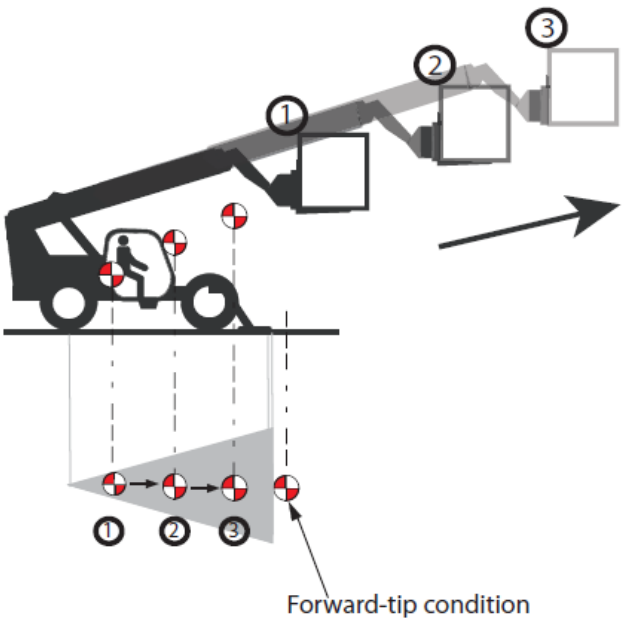
Extending the boom

Extending the boom of the telehandler when the machine is loaded causes the center of gravity to move higher and toward the front of the machine.

The heavier the load being lifted, the further the combined center of gravity will move toward the front of the stability triangle.

If the combined center of gravity moves beyond the front of the stability triangle due to lifting a load in excess of the machine's rated capacity, a *forward tip-over* will occur.

Always refer to the machine's _____ _____ and never exceed the machine's rated lift capacity in any of the lifting zones.



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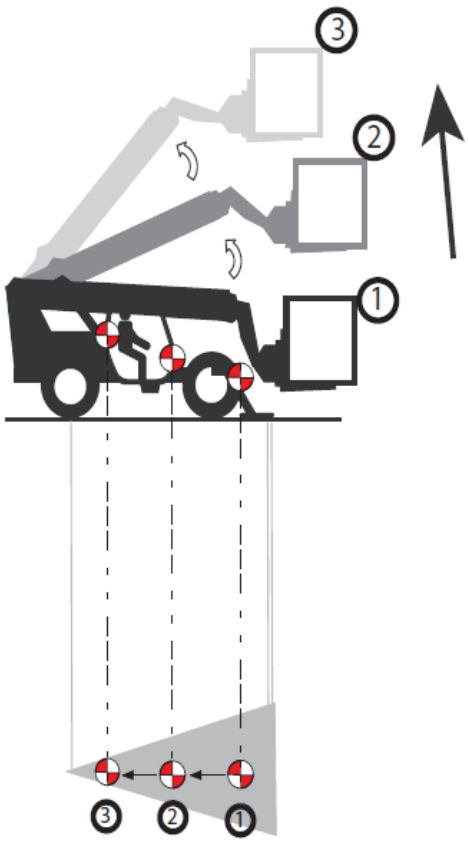
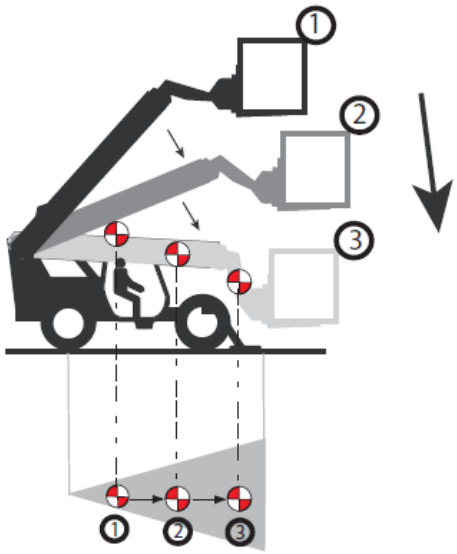
Raising the retracted boom

Raising the retracted boom of the telehandler, with or without the load, moves the center of gravity higher, away from the ground, and toward the rear of the machine.



Lowering the retracted boom

Lowering the retracted boom of the telehandler, with or without the load, moves the center of gravity lower and toward the front of the machine.



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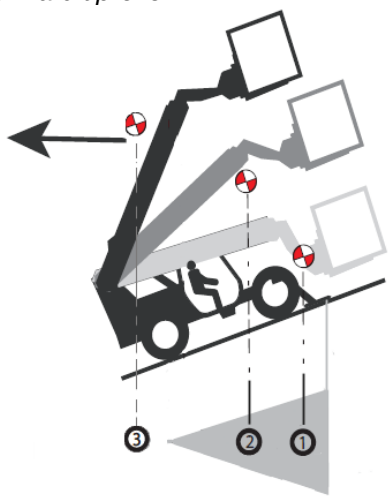
Moving the boom while perpendicular to a Slope

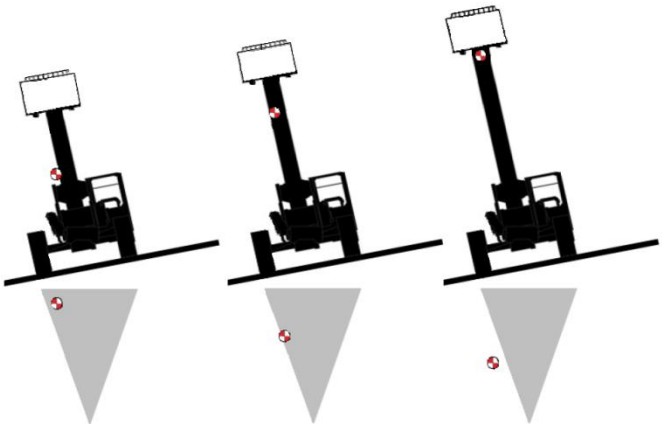
The telehandler boom should **never** be raised if the machine is not level.

If the boom is raised to a high boom angle while the telehandler is on a slope, the center of gravity will shift rearwards and out of the stability triangle and this will cause a *backward tip-over*.



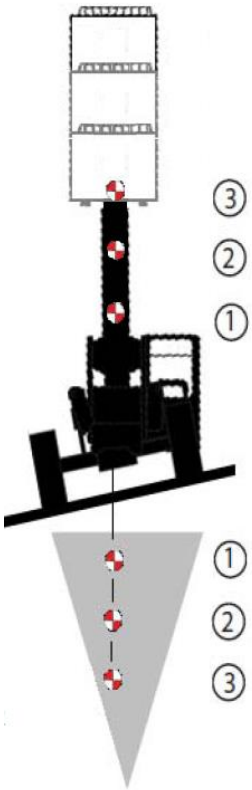
Raising the boom while parallel to a slope

If the boom is being raised while the machine is on a side slope or not level, the center of gravity moves higher and toward the rear of the telehandler.



As the boom continues to be raised, the center of gravity position may fall outside the stability triangle and this will cause *a side tip-over.*

Always use the frame sway control to level the machine on a slope _____ raising the boom to improve machine stability.



Always make sure that ground conditions are adequate to support the load forces imposed by the machine and the load. Do not use the frame sway feature when operating on unstable ground or slippery surfaces.

Do not raise the boom if the machine is not level.

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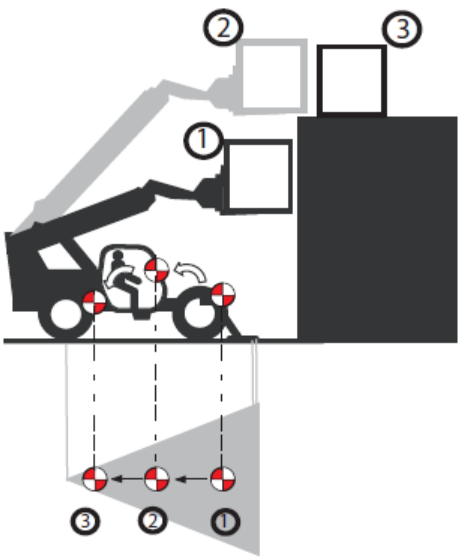
Unloading when raised

When the telehandler forks are loaded and the boom is raised from the stowed position, the center of gravity moves higher and towards the rear of the machine.

Unloading the telehandler when the boom is raised moves the center of gravity lower and toward the rear of the machine.

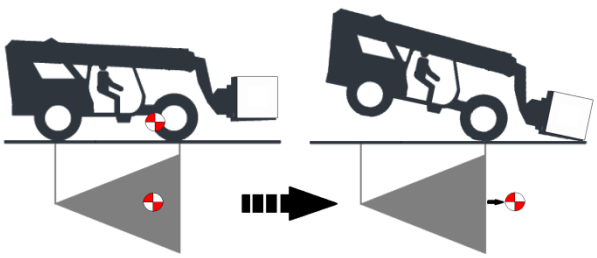


Dynamic forces

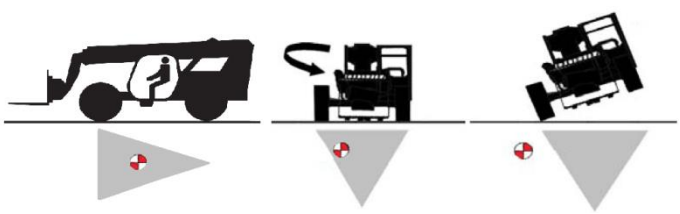
Telehandler operation, such as machine motion or moving suspended loads, creates dynamic forces on the machine and the load. Dynamic forces can suddenly shift the _____ of the combined center of gravity to outside of the stability triangle and cause the machine to become unstable.

Some conditions which may cause instability due to dynamic forces are:

Heavy braking: When a telehandler is traveling forward with a load and heavy braking is applied, the change in dynamic forces results in a sudden forward shift in the load's center of gravity. If this happens, the load may fall from the telehandler. If the load is fixed, the *combined* center of gravity may move outside the stability triangle, which could cause a *forward tip-over*.



High-speed turning: Turning the telehandler, with or without a load, causes lateral dynamic force which can result in a sideways shift in the center of gravity. If the telehandler turning speed is too high, the center of gravity may move outside the stability triangle, which could cause a *side tip-over.*



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Stability condition when the machine is equipped with a rear axle lock system

Some telehandlers are equipped with a rear axle lock system which restricts certain telehandler operations in order to minimize the likelihood of the operator reaching a condition of instability in certain situations.

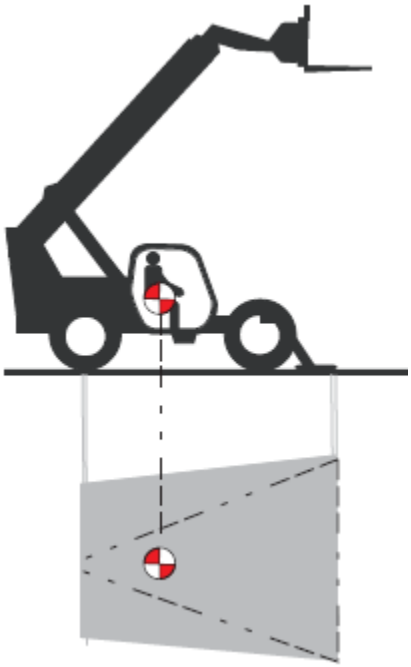
The system will typically engage at a predetermined boom elevation and boom extension. When the system engages, among other things, it can lock out the travel controls and engage the parking brake system so that you are not able to drive.

When the operator returns the telehandler to the stowed operating position, disabled functions then become active.

When the rear axle lock is locked into a fixed position, the stability zone for the telehandler changes from a triangle to a rectangle on machines not equipped with stabilizers, or to a trapezoid on machines equipped with stabilizers, increasing lateral, or side-to-side, stability.

Note: The rear axle locking condition does not increase longitudinal, or front to back, stability.

See the Operator's Manual for a complete description of the rear axle lock system.



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Controls

Steering

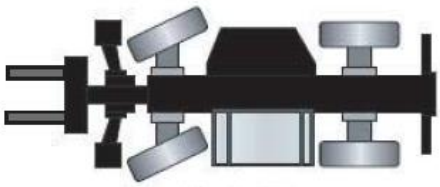
In most telehandlers, operators can steer the machine using multiple steering modes such as two-wheel, four-wheel coordinated and four-wheel crab modes. Each mode provides specific maneuverability characteristics that are advantageous in certain conditions.

For example:

Two-wheeled steer is appropriate for over the road travel or when navigating a large jobsite.

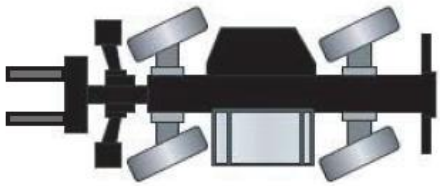
Four-wheeled coordinated steer provides a much tighter turning radius for navigating a small or congested jobsite.

Four-wheeled crab steer allows the operator to move the machine sideways for easier load pickup and placement or when navigating within a tight area.



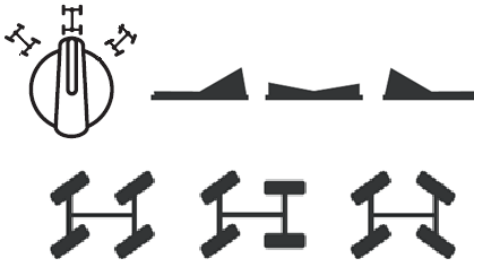
Two Wheeled Steer

Four Wheeled Steer



Crab Steer

The steering selection switch is usually a three-way switch, as shown in the figures below. The switch should clearly indicate the steering mode using visual icons.



Note: Always realign the wheels before shifting to a different steering mode

Instrument Panel

The instrument panel allows the operator to constantly monitor specific machine conditions and performance. Machine operation parameters such as engine RPM, fuel level, water temperature, hour meter, and steering mode are typically displayed on the instrument panel located inside the operator's cabin. Machine condition and warning lights for battery condition, low fuel level, and parking brake may also be included in the instrument panel.

Different manufacturers may have gauges and warning lights placed in different locations. Please refer to the manufacturer's Operator's Manual for detailed information on the specific machine's instrument panel.

Drive controls

Telehandlers are typically equipped with a powershift or hydrostatic transmission. A steering wheel provides directional control of the telehandler. Using the steering wheel, in conjunction with the transmission controls, the operator can control the machine's travel speed and direction according to jobsite conditions.

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Control Joysticks

The telehandler is typically equipped with either a single or multiple control joysticks.

Control joysticks allow the operator to control boom and attachment functions, and sometimes frame sway functions. Boom functions include extending, retracting, raising and lowering of the boom. Attachment movements include tilting forward and backwards.

Auxiliary hydraulics controlling optional powered attachments may also be controlled through the joysticks.

Auxiliary Hydraulics

Some telehandlers may be equipped with auxiliary hydraulics that allow the operator to connect and operate hydraulically-powered attachments that allow them to perform specific tasks at the work location.

Manufacturers will provide controls to operate the auxiliary hydraulic system in the cab. Refer to the machine's Operator's Manual for more information.

Axle Differential Lock

Some telehandlers are equipped with an axle differential lock that improves traction, by locking the drive wheels together in certain types of terrain such as sand, mud or snow.

Typically, a floor mounted, foot operated pushbutton switch or panel mounted switch is used for engaging the axle differential lock.

Always come to a _____ _____ before activating the axle differential lock. Do not engage the differential lock when the machine is moving, or when performing work on dry pavement, or significant machine damage may occur.

Parking Brake

The parking brake should always be applied _____ picking or placing a load or leaving the machine, in order to prevent unintended travel.

Typically, a panel mounted switch inside the operator’s cabin engages and disengages the parking brake.

Service Brake

A floor mounted service brake is used to control the machine speed and to stop the machine motion.

Typically, the service brake is controlled using a foot pedal. Push and hold the service brake pedal to stop the machine. Use controlled foot pressure during travel to control the machine speed.

Accelerator Pedal

A floor mounted, foot operated accelerator pedal is also used to control the speed of the telehandler. Press down on the accelerator pedal to increase speed and release the accelerator pedal to decrease speed.

The accelerator pedal can also be used to increase the pump flow to the machine's hydraulic system. However, always engage the parking brake before doing this to prevent unintended machine movement.

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Frame Sway

Some telehandlers have a system that can level the chassis to the ground. The purpose of this system is to help maintain the stability of the telehandler on uneven terrain. This system is often referred to as the Frame Sway or Chassis Leveling control.

This allows the operator to level the chassis of the telehandler to the right or left. The range varies, but typically this leveling capability can range between 8- and 12-degrees rotation.

This system should _____ be operated when the boom is elevated above horizontal, and should only be used to help level the telehandler _____ raising the boom

Stabilizers

Some telehandlers may be equipped with stabilizers at the front of the machine. Stabilizers help maintain the stability of the machine during lifting of the boom or load and can in many cases increase the lifting capacity of the telehandler in specified load zones.

Stabilizers may be permanently fixed to either the chassis frame or the front axle.

Do not _____the stabilizers with an elevated boom and do not use the stabilizers to level the telehandler. If needed, fully lower and retract the boom or load, then reposition the stabilizers.

Frame mounted stabilizers may be affected by the use of the frame sway controls. Check with the telehandler manufacturer to determine if frame leveling should be done before deploying stabilizers.

When using stabilizers, the operator must use the correct load chart for verification of capacity limits within the various load chart zones.

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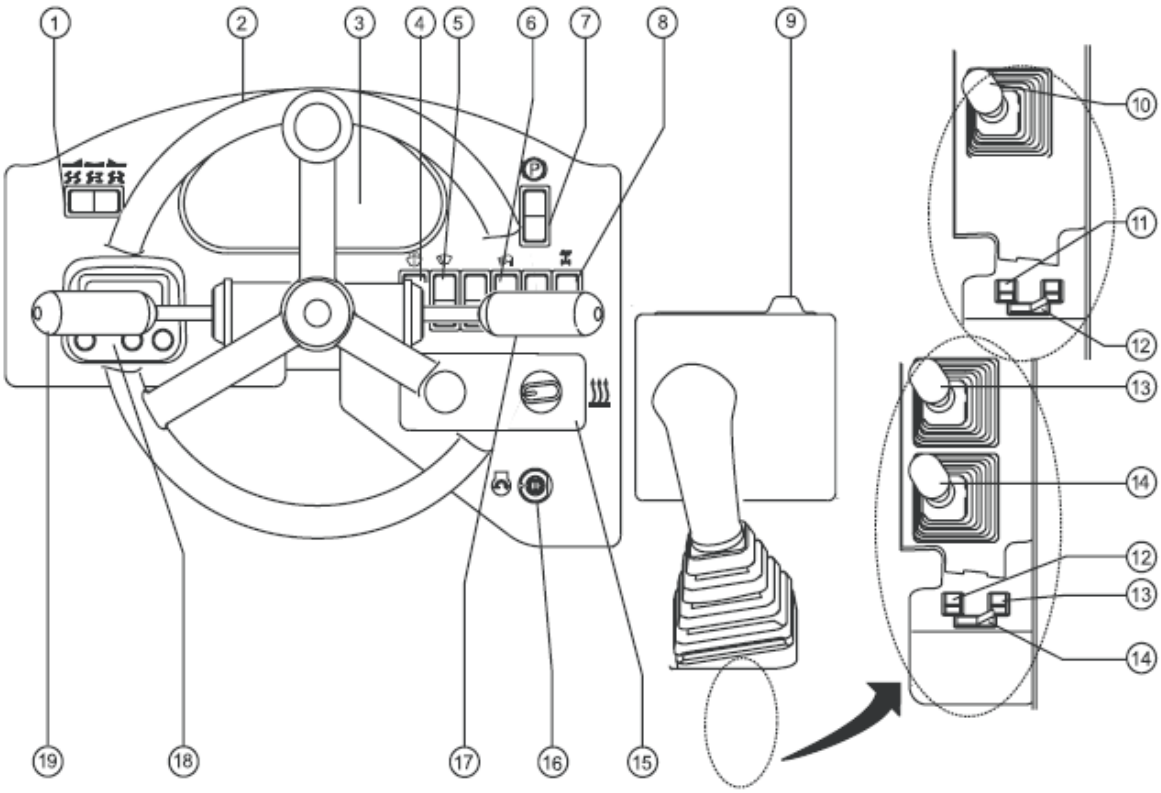
Dashboard Controls Example



1. Steer Select Switch
2. Steering Wheel
3. Instrument Panel
4. Windshield Wiper Switch (If equipped)
5. Windshield Wiper Switch (If equipped)
6. Work light switch (if equipped)
7. Parking brake switch
8. Axle differential lock switch(if equipped with dashboard switch)
9. Load chart holder
10. Control joystick (Single control handle option)
11. Stabilizer switch
12. Auxiliary hydraulic switch
13. Boom control handle (dual control option)
14. Tilt and sway control handle (dual control option)
15. Heater controls (if equipped)
16. Ignition switch
17. Road light control
18. Engine condition indicators
19. Transmission control

NOTE: For detailed information on machine operation and controls, please refer to the operator's manual.

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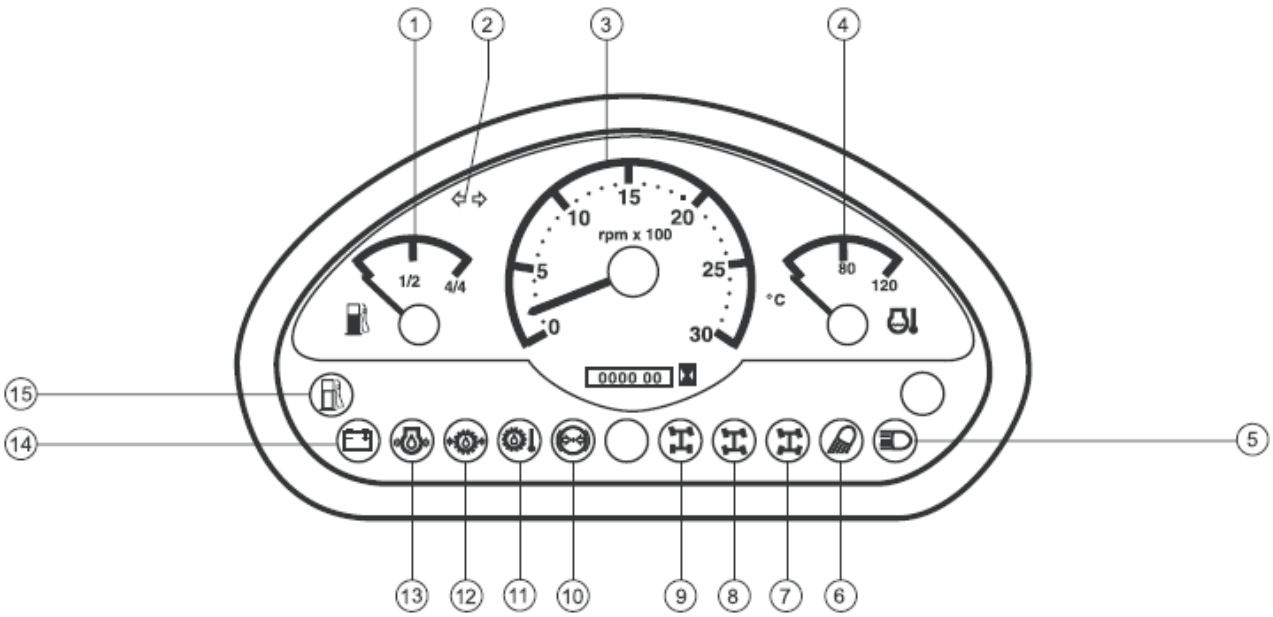
Instrument Panel Controls Example



1. Fuel level gauge
2. Turn signal indicator lights
3. Tachometer and hour meter
4. Engine coolant temperature gauge
5. Driving lights indicator lights
6. Working lights indicator lights
7. Four-wheel steer indicator lights
8. Crab steer indicator lights
9. Two wheel steer indicator light
10. Parking brake indicator light
11. Transmission oil temperature indicator light
12. Transmission oil pressure indicator light
13. Engine oil pressure indicator light
14. Battery condition indicator light
15. Low fuel level indicator light

NOTE: For detailed information on instrument panel please refer to the operator's manual.

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Telehandler Operation

Startup

1. Enter the cab using a _____-_____ entry, facing the machine and using the steps and handrails provided.

2. Adjust the seat and steering wheel until you can easily reach all the machine controls. Do not operate the controls from any place other than the seated position.

3. Fasten the seatbelt. Remember to always wear the seatbelt while operating the telehandler.

4. Adjust side and cab mirrors.

5. If the controls are not already in the neutral position, set them to the neutral position.

6. If the parking brake is not already set, set the parking brake.

7. Start the engine.

8. Monitor the gauges and the instrument panel to ensure that the telehandler is working properly.

Traveling with or without a load

Driving conditions on job sites and roads will frequently change.

Operators should be properly trained and use their best judgment and _____ to constantly assess the jobsite conditions and avoid exceeding the machine's, or the operator's, capabilities for the terrain and conditions.

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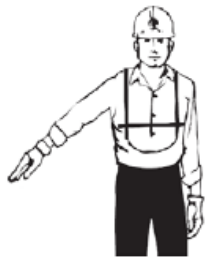
Hand Signals

To ensure the proper and safe handling of loads and machine movement in the workplace, hand signals are sometimes used to assist the telehandler operator. It is critical for the operator and ground personnel to agree upon and understand all hand signals used on the jobsite before beginning the operation.

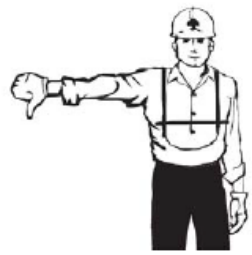
Some examples of hand signals used for telehandler operation are shown below. Please consult your employer if these signals are different than those being used for a specific jobsite or task.

TILT FORKS UP
With one arm held at side, extend other arm upward at about a 45 degree angle.



TILT FORKS DOWN
With one arm held at side, extend other arm downward at about a 45 degree angle.

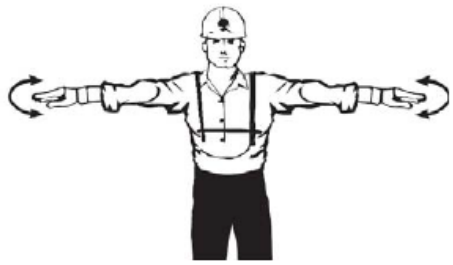
RAISE BOOM
Arm extended, fingers closed, thumb pointing upward.



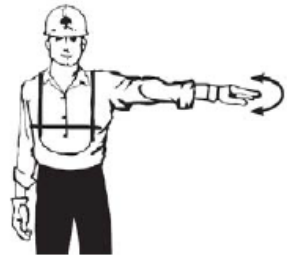
LOWER BOOM
Arm extended, fingers closed, thumb pointing down.

EXTEND BOOM
Hold both fists in front of body with thumbs pointing outward.

RETRACT BOOM
Both fists in front of body with thumbs pointing toward each other.



EMERGENCY STOP
Arms extended, palms down, move hands rapidly back and forth.



STOP
Arm extended, palm down, move hand back and forth.

TRAVEL
Arm extended forward, hand open and slightly raised, pushing motion in direction of travel.

THIS FAR TO GO
With hands raised and open inward, move hands laterally, indicating distance to go.

MOVE SLOWLY
One hand gives motion signal, Other hand motionless in front of hand giving the motion signal.

STOP ENGINE
Draw thumb or forefinger across throat.

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General operator guidelines during traveling

All traffic regulations must be observed by the operator including but not limited to:

- While traveling, with or without the load, the operator must speed up or slow down gradually, in a_____, safe manner, to prevent the load from shifting or the telehandler from potentially tipping over.
- Reduce speeds before negotiating turns.
- Always observe weight restrictions for surfaces intended to hold the load.
- Keep a clear view of the path of travel.
- Ensure that the path of the telehandler is clear of pedestrians and obstructions.
- The operator must be aware of the steering mode being used on the equipment.
- In certain cases, chances of lateral overturning are greater when traveling _____ a load. The operator must maneuver at a slow, controlled speed, even when traveling without a load.
- Keep to the right side of path of travel. If visibility is obstructed always use a spotter who is familiar with the proper hand signals.
- Follow speed limits permitted for the work area.
- Safe distance (_____ truck lengths) must be maintained between other telehandlers or other construction equipment.
- Slow down on wet, slippery roads and surfaces.
- Avoid driving over loose objects or surfaces in the path of travel.
- Avoid all other potential hazards.

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Guidelines during traveling cont.

- Right of way must be yielded to emergency vehicles, such as ambulances or fire trucks.
- When approaching an intersection or when the operator's view is obstructed during operation, the operator is required to_____, sound the horn and then proceed with caution.
- Drive in the reverse direction (traveling with load trailing) in cases when the forward view is being obstructed.
- Whenever backing the telehandler from an obstruction, use proper backup procedures and be aware of any blind spots. Use the mirrors, backup alarm and horn. If blind spots exist, use additional personnel to help navigate around them or to warn others.
- Use proper load-engaging techniques such as keeping the load height low and the forks tilted backward to secure the load while traveling.

Driving in confined areas:

In a confined area such as in a narrow aisle or between narrowly spaced buildings, remember to:

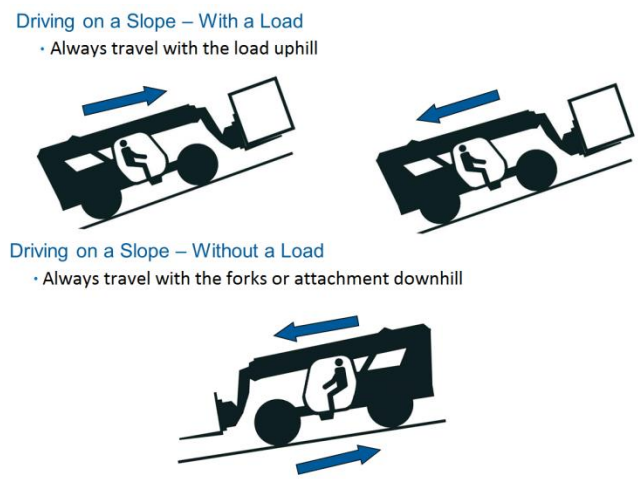
- Steer very carefully
- Ensure that the load does not exceed the width of the space available
- Watch for low doorways
- Ensure that there is adequate lighting and ventilation
- Exercise caution whenever entering or leaving a building on bright sunny days. Allow time for your eyes to adjust to the change in light before proceeding.

Driving on a slope:

Operators must use caution while driving on slopes. Grades should be ascended or descended slowly.

When the machine is loaded, the operator must always travel with the load_____.

When the machine is unloaded, travel with the forks or attachment_____.



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Driving on a slope cont.

Always travel perpendicular, or straight on, to the slope and always keep the machine in gear. Do not turn across a slope. Limit travel path and speed according to the condition of the ground surface, traction, slope, location of personnel and any other factors which may create a hazard.

Never drive the machine unless the telehandler and the load are in their proper travel position.

Procedure to follow during tip-over

During a tip over, the operator should remain in the cab and brace for impact by grasping the steering wheel and moving the body _____ to the direction of tipping.

All body parts should remain inside the operator's cabin at all times.

- Do not attempt to jump clear of the machine as this may result in serious injury or even death.
- The seat belt must always be worn during machine operation. If the machine tips over while failing to wear a seat belt, serious injury or even death may occur.

Pedestrians

Always yield right-of-way to pedestrians.
The telehandler operator must take care when working in areas with pedestrians or personnel, so as to not cause any bodily harm or injury. Use caution when approaching pedestrian areas. Before and during traveling, always look around for any pedestrians.

No person should be allowed to _____ or_____under the elevated portion of the telehandler forks or attachment, whether the telehandler is unloaded or loaded.

Elevators

The operator must ensure that the elevator can support the telehandler and its load. Approach elevators slowly and ensure that the elevator is properly leveled with the building floor.

There should be adequate clearance for the telehandler to enter or leave the elevator.

Enter the elevator squarely with load end in_____. Once the telehandler has entered the elevator, set the brakes and follow proper shut-down procedures.

Docks, trailers and railcars

In cases when a telehandler is operated on dock boards, bridge plates or rail cars, precautionary steps must be taken by the operator.

Dock boards:

- Dock boards should be properly secured to prevent accidental movement during operation.
- Dock boards should have a high friction surface for good grip.
- All dock boards should have proper markings that mention capacity rating.
- Dock boards should have substantial contact with the dock or loading platform and the transport vehicle or rail car to prevent rocking or sliding.
- All surfaces that the telehandler will travel over must be capable of supporting the combined weight of the telehandler and the load.

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Transporting on trailers:

- Brakes and wheel chocks should be applied to trucks before being loaded or unloaded with telehandlers.

Railcars:

- Railcars should be secured with brakes, wheel stops or other recognized positive means before the loading operation.

- Do not use forks to open or close railcar doors unless the truck utilizes a device specifically designed for opening and closing railroad car doors and the operator is trained in its use.

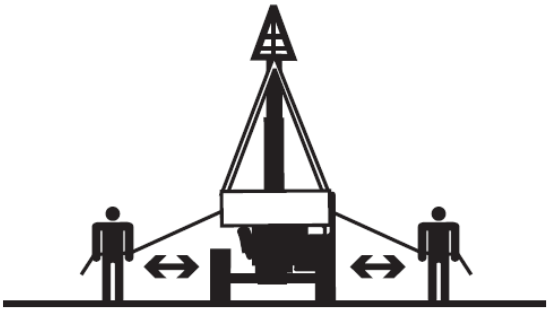
Suspended Loads

Refer to the manufacturer's specifications and Operator's Manual on safe handling of suspended loads.

Whenever a telehandler is intended to be used to lift and transport a suspended load, the operator must be _____ and qualified in identifying and avoiding the potential hazards in handling a suspended load on the site.

Telehandler load charts are designed for loads where the load center is stationary. As a suspended load is moved, the load center can _____. As a result, the standard load chart does not apply and extreme caution in transporting and lifting the load should be observed.

Only transport suspended loads with approved attachments by the telehandler manufacturer and do not transport suspended loads with chains or slings over the forks or carriage unless authorized by the telehandler



The handling of suspended loads by means of a boom or other device can introduce dynamic forces affecting the stability of a rough terrain forklift truck that are not considered in the stability criteria of ANSI/ITSDF B56.6 Safety Standards.

Grades and sudden starts, stops, and turns can cause the load to swing and create a hazard if not externally stabilized.

If lifting and transporting a suspended load is permitted on the job site, the following criteria should be followed:

- The user and the operator shall comply with all applicable jobsite, local, state, provincial, or federal rules, regulations, directives and standards related to the use of the truck with a suspended load.

- The user shall ensure that the operating personnel are given the necessary training.

When handling suspended loads:

- Speed must be limited by any conditions that could cause any unexpected movement of the load, or jeopardize the safe transport of the load

- Do not exceed walking speed

- During transport and lifting, use guide ropes or tag lines by qualified personnel to help control the load and prevent it from swinging

- When visibility is or could be obstructed in the direction of travel, the operator must use alternative or additional means to safely transport the load, such as a spotter or a qualified signal person

- Transport the load with the bottom of the load and the boom as low as possible

- Only travel on solid surfaces

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Raising and placing a load

The load chart in the cab shows the capacity limits of the telehandler. As mentioned previously, to use the load chart, the operator must know the _____ of the load and how far out and up it is to be placed.

A telehandler may have more than one load chart. Be sure you are using the appropriate load chart that corresponds to the specific attachment and the telehandler model.

Procedure:

1. Travel to the desired final destination on the jobsite where you will be placing the load. Bring the telehandler to a gradual stop.

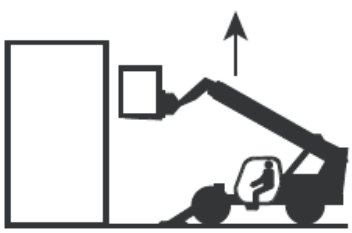
2. Place the transmission in neutral.

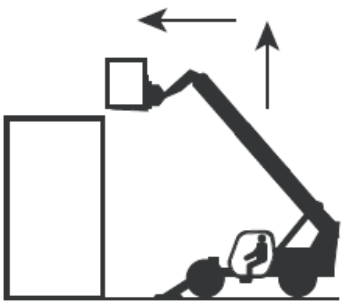
3. Apply the parking brake.

4. Lower the stabilizers, if equipped and if they are required for additional capacity.

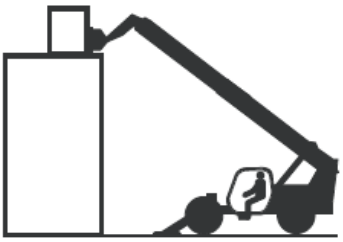
5. Level the chassis, if equipped, with the leveling feature.

6. Gradually move the controller to raise the boom to the required height.

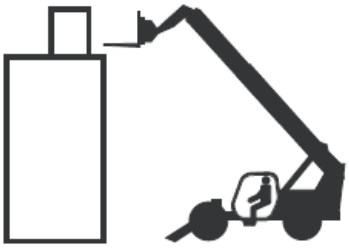




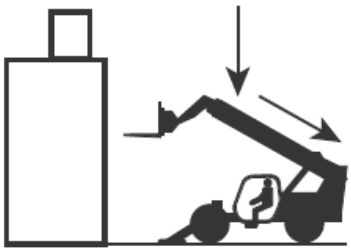
7. Gradually move the controller to land the load at the final position. Lower the load until the weight s completely off the forks. Do not apply a downward force with the forks.



8. Gradually move the controller to bring the forks out of the load.



9. When the forks are clear of the load and the structure, the boom can be retracted, then lowered.



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Load retrieving with forks

1. Position the telehandler perpendicular with the load.

2. Spread the forks as wide as possible while still being able to pick up the load.

3. Level the forks to align with the entry point of the load.

4. Raise the forks to the proper load entry height.

5. Gradually move the controller to place the forks under the load.

6. Slightly raise the load so it is completely lifted free from its resting point.

7. Tilt the load back so that the fork carriage is supporting the load.

8. If picking from a ground position, carefully pick up the load, raise it to travel position and keep the forks tilted back slightly to cradle the load.

9. If lowering from an elevated position, completely lower the load by first retracting then lowering the boom to travel position.

Stacking and unstacking

Depending on workplace rules and regulations, an operator may be allowed to stack or unstack loads.

Beware of stack height limitations. Avoid over-stacking, as it may crush lower levels of loads or could cause the stack to become unstable.

While stacking, place loads only on a_____surface, which can sustain the weight of stacked loads. During unstacking, be aware of the clearance around the load and working area.

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General

Refueling

Follow shut down procedures before refueling the telehandler. Never re-fuel if the engine is still running. Open flames and smoking are not permitted in proximity of refueling areas.

After refueling, tighten the fuel tank cap before starting the engine.

In case of spillage, ensure that the correct spill procedures are followed. Contact your local health and safety representative for more information.

Recharging Batteries

Follow shut down procedures before recharging the battery.

Open flames and smoking are not permitted in proximity of recharging areas.

Batteries contain acid. Avoid spilling or contacting battery acid. Neutralize battery acid spills with baking soda and water.

Ensure there is adequate ventilation for the dispersal of fumes from gassing batteries.

Shutdown

Telehandlers should be parked in_____areas. The parking location should be located away from potential hazards such as slopes, traffic flow exits, pedestrian walkways, elevators, ramps or fire-fighting equipment.

Follow shutdown procedures when the machine is left in an unattended state.

Note: A telehandler is considered unattended when –

- The operator is 25 ft. or more away from the telehandler even if it is in view of the operator.
- Whenever the telehandler is not in the view of the operator.

Procedure for parking the unit and shutting the machine down:

1. Park the telehandler in a safe location.
2. Bring the telehandler to a complete stop.
3. Apply the parking brake.
4. Fully retract the boom and lower the attachment. If equipped with forks, lower the forks flat on the ground to avoid a trip hazard.
5. Place all directional controls in neutral.
6. Turn off the telehandler.
7. Cycle the hydraulics to relieve any remaining pressure.
8. Remove the key.
9. If there is a separate fuel source, turn it off.
10. Dismount facing the telehandler using the 3-point contact method described earlier.
11. If the telehandler must be left on an incline, park at a right angle to the slope.

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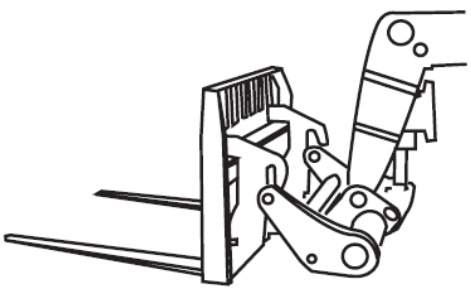
Attachments

There are a wide variety of attachments available for telehandlers. As mentioned previously, only use attachments that have been _____ by the telehandler manufacturer. This will help ensure that the proper load charts, and any conditional requirements, are provided to the operator.

Some attachments may be powered by the auxiliary hydraulic system on the telehandler.

Attachments may be permanently fixed to the telehandler or utilize what is often called a “quick- attach” system that allows the telehandler to quickly change different attachments.



Example: Quick-Attach™ System

Please note that telehandler attachments may:

1. Affect the _____ of the telehandler
2. Create a “Partially Loaded Condition”
3. Change the load zone_____.

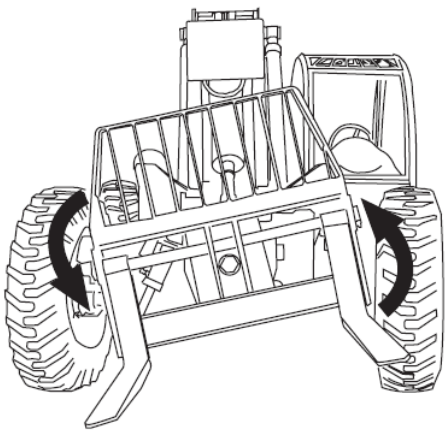
Please refer to the Operator’s Manual for more information on using attachments or contact the telehandler manufacturer.

Shifting and Rotating

A shifting attachment can shift an entire load by shifting its fork carriage or forks to the right or left of the boom. The purpose of this attachment is to help improve the ability to precisely position the forks or carriage to pick or place a load.

A rotating carriage allows the operator to rotate the entire carriage, in many cases up to 10 degrees, clockwise or counterclockwise. The purpose of this attachment is to allow the operator to compensate for uneven load surface or uneven terrain when it affects an area that a load needs to be picked or placed.

Note: Never use the frame sway function to level an elevated load.



Example: Rotating Carriage

Swing Carriage

Swing carriages are typically used for helping land a load in tight areas, or maneuvering loads during travel around tight corners. Swing carriages allow the operator to move the carriage clockwise or counterclockwise along the horizontal axis.

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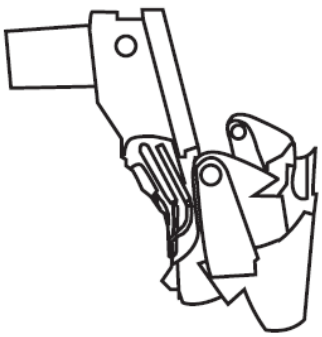
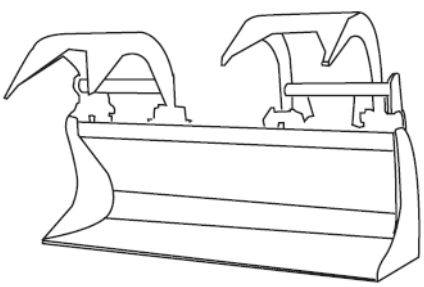
Universal Attachment Adapter

The purpose of this attachment is to allow the telehandler to use a wider variety of attachments. As a result, the versatility and utilization of the telehandler can be improved.



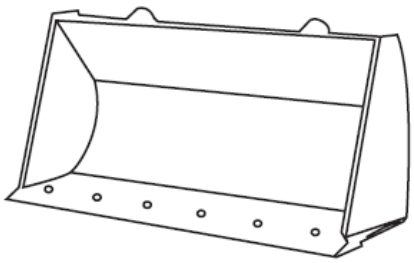
Industrial Grapple Bucket

Grapple buckets are intended to be used to move large or heavy pieces of material. It is a good solution for jobsite cleanup or moving logs or boulders.



Heavy-Duty Multi-Purpose Bucket

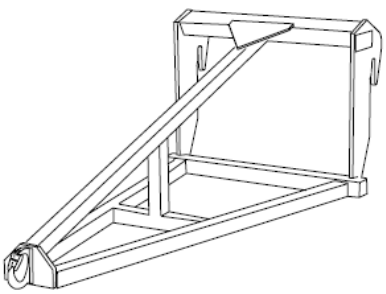
Multi-purpose buckets are only approved for certain telehandlers. Since they subject the telehandlers boom to more torsional stress and impact loading, not all telehandlers are suited for this type of attachment. If unclear whether this attachment is approved for a specific telehandler, contact the telehandler manufacturer.



Truss Boom

Truss booms are often used for transporting large or hard to handle loads, such as large wooden trusses. Truss booms are almost always rated for less than the overall capacity of the telehandler. Always use the appropriate load chart for a truss boom to avoid overloading.

Some truss booms have multiple positions to extend a load. The rated carrying capacity of the truss boom may be reduced as the position of the load extends further out from the machine.



Note: Refer to the operator's manual for information on handling suspended loads.

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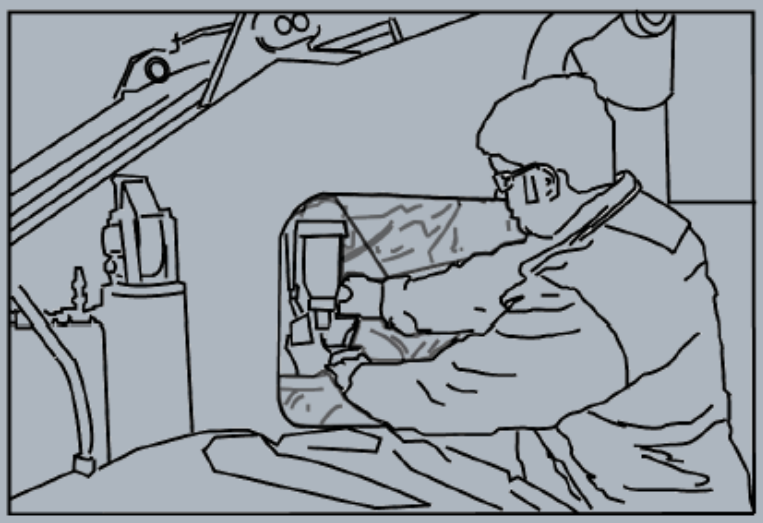
MAINTENANCE



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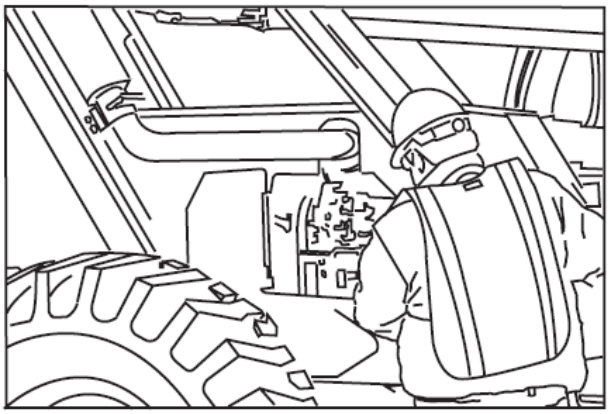
Maintenance



If at any time a telehandler is found to be in need of repair or unsafe for operation, the telehandler should be immediately _____and _____ from service until it can be repaired by a qualified service technician, according to the manufacturer's specifications.

Only routine maintenance items specified in the operator's manual and approved by the employer may be performed by the operator.

Routine Maintenance Items

Only perform routine maintenance when the engine is turned off. Allow the engine to cool before accessing areas under the engine or radiator covers.

Scheduled maintenance inspections, above and beyond routine maintenance, should only be completed by _____ service technicians, according to the manufacturer's specifications.

Examples of Some Routine Maintenance Items are:

• Check the engine oil

• Check the engine coolant

• Check the hydraulic oil

• Check the tire pressure

• Check the batteries

For additional routine maintenance-specific information, refer to the Operator's Manual.

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Appendix

Standards Compliance

Many aspects of telehandler operation and testing are discussed in standards and regulations published by the regulatory and standards institutions.

Institutions responsible for publishing standards by country are mentioned below.

U.S.A. - The American National Standards Institute (ANSI), the Industrial Truck Standards Development Foundation (ITSDF) and the Occupational Safety & Health Administration (OSHA). For class 7 variable reach trucks (telehandlers), standards pertaining to operator training are also found in ANSI/ITSDF B56.6 and OSHA 1910.178(l).

Canada – The Canada Standards Association (CSA) and the Canadian Occupational Health and Safety (OHS). For class 7 variable reach trucks (telehandlers), standards pertaining to operator training can be found in CSA B335. Please refer to specific OHS requirements for your province.

Details on above mentioned standards can be found at the links provided below.

ANSI/ITSDF - www.itsdf.org

CSA - www.csa.org

OSHA - www.osha.gov

OHS - www.ccohs.ca

NOTE: Equipment standards can be revised. Check often for any updates.

Resources

In addition to this training manual, you have additional access to product information resources for acquiring information related to the design and proper use of the equipment.

Equipment dealers are a great resource for additional safety training materials. They can provide specific Operators and Safety Manuals for a specific type of telehandler being used as well as offer operator safety training.

In addition to the previously mentioned standards and regulatory websites, you might also consider visiting the equipment manufacturer’s website for more information. In many cases, as is the case with Genie’s website, you can find the necessary Operators Manual for the equipment. If you are unable to find the applicable manual, you can also contact the manufacturer directly.

Ordering Training Materials

To order replacement training materials for this program please contact your local Genie dealer. To assist you in finding a local Genie dealer in your area please visit us at www.genielift.com.

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Written Exam

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Genie LIFT PRO™

TELEHANDLER OPERATOR TRAINING

Class 7 Rough Terrain Variable Reach Forklifts Operators Safety Training

WRITTEN TEST

Name: _____ Date: _____

1. It is not required by the Occupational Safety and Health Administration that operators receive formal and hands-on training prior to operating a rough terrain forklift.

a) TRUE
b) FALSE

2. It is the responsibility of the _____ that each rough terrain forklift operator is competent and certified to operate a powered industrial truck safely.

a) government
b) operator
c) employer
d) manufacturer

3. Safe use of the telehandler and following worksite rules is the responsibility of the _____.

a) government
b) operator
c) employer
d) manufacturer

4. The main reasons why operator training is important:

a) The safety of operator and bystanders
b) To protect company property
c) It's the law
d) All of the above

5. The term FOPS for a telehandler refers to:

a) Falling Object Protective Structure
b) Fixed Operator Position Symbol
c) Free Object from Pallet System
d) Force On Pull Structure

6. The operator's manual _____.

a) is required to be read by the operator prior to operating the rough terrain forklift.
b) is typically located inside the operator's cabin
c) can be different for different machine models
d) All of the above

7. Which safety symbol has the highest level of hazard or danger associated with it?

a) Warning
b) Notice
c) Danger
d) Caution

8. Which is not a characteristic of a professional operator?

a) Responsible
b) Wears personal protective equipment
c) Capable of moving a load faster than anyone else
d) Keeps the rough terrain forklift under control at all times

9. Completing this telehandler operator training course allows the operator to operate every model of telehandler without familiarization.

a) TRUE
b) FALSE

10. Before operating the rough terrain forklift, operators are not required to:

a) inspect the workplace
b) perform pre-operation inspection
c) perform function tests
d) none of the above

11. Which is the best load handling technique?

a) Drive with the load raised for better visibility
b) Retract boom and then lower load
c) Lower load with boom extended
d) None of the above

12. When raising the load, the operator should _____.

a) use sway controls to position the elevated load
b) ensure machine is level before raising the load
c) simultaneously drive to unloading position
d) avoid using the parking brake

13. When approaching an intersection or when visibility is restricted during traveling (such as traveling upwards with a load on a slope), operator should _____.

a) honk to alert pedestrians and continue driving
b) honk several times and drive very quickly
c) stop, sound the horn and proceed with caution
d) slow down and check for pedestrians

14. Which one of the following is <u>not</u> a tip-over hazard?

a) Operating within the specified limits of the machine found in the operator's manual
b) Gusty wind conditions
c) Overinflated or damaged tires
d) Uneven terrain, debris, slippery surfaces or unseen drop-offs

15. Which is a true statement?

a) Authorized personnel are allowed under an elevated boom
b) Personnel should wear hard hats when working under an elevated load
c) No person is allowed under an elevated boom
d) Personnel can be transported using telehandlers

16. In order to safely enter the telehandler,

a) use the steering wheel to pull yourself into the cab
b) use three-point contact method by utilizing the cab steps and hand-rails
c) use door handle during entry
d) use any technique mentioned above

17. Escaping hydraulic fluid or air under pressure can penetrate skin, causing serious injury, which may lead to long-term serious injuries if not treated immediately.

a) TRUE
b) FALSE

18. Which statement is true about using telehandlers near power lines?

a) Telehandlers are not electrically insulated
b) Maintain10 ft. clearance for all supply voltages
c) Operator cab is electrically insulated
d) Safe operating distance depends on the supply voltage - refer to Required Clearance information in the operator's manual and on decals on the machine.
e) Both (a) and (d)

19. Pre-operation inspection _____.

a) is a visual inspection done prior to each work shift
b) is done to test all machine functions
c) is completed once, prior to the first operator shift
d) is completed after function test and before operating the machine

20. The operator is required to always wear the seatbelt when using the machine.

a) TRUE
b) FALSE

21. Select the option which mentions two differences between automobiles and rough terrain forklifts.

a) Four wheels and seatbelts
b) Engine and steering
c) Center of gravity and weight
d) Brakes and control layout

22. It is generally okay to modify a telehandler or it's attachment without the manufacturer's prior written approval.

a) TRUE
b) FALSE

23. Load charts _____.

a) are only needed during pre-operation inspection
b) indicate engine rpm requirements for lifting the load
c) indicate maximum weight of telehandler
d) indicate the maximum machine capacity in different lifting zones, and in different machine configurations.

24. The operator may slightly exceed the stated load chart capacity as all manufacturers add a safety factor into machine capacity calculations.

a) TRUE
b) FALSE

25. Every unique machine model/attachment combination has a unique load chart diagram.

a) TRUE
b) FALSE

26. The center of gravity is _____.

a) the single point on an object at which all of the object's weight is concentrated.
b) the single point where gravity forces on the load are the strongest
c) the point mid-way between the center of an object and the ground
d) none of the above

27. On a telehandler that is not equipped with stabilizers, which is <u>not</u> one of the three supporting points of a stability triangle?

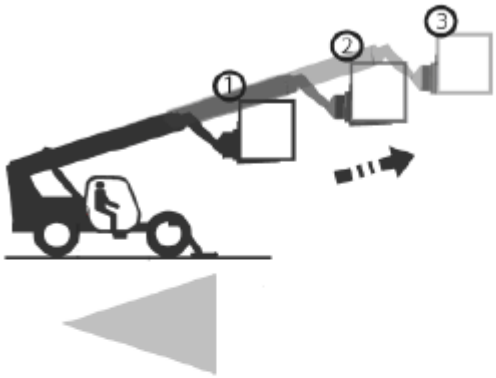
a) Center point on the rear axle
b) Left front tire
c) Left rear tire
d) Right front tire

28. What happens when the combined center of gravity moves out of the stability triangle?

a) Loss of steering
b) Loss of traction
c) Tip over
d) Rear axle locks

29. In certain cases, the potential of a lateral tip-over are greater when traveling without a load.

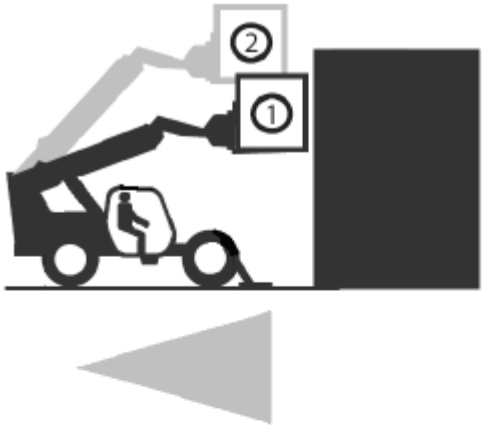
a) TRUE
b) FALSE



30. For the figure shown above (when the load is raised and extended), which best describes the correct displacement of the combined center of gravity of the machine?

Combined center of gravity (CG) moves _____.

a) lower and towards the front of machine
b) higher and towards the front of machine
c) lower and towards rear of machine
d) higher and towards rear of machine



31. For the figure shown above (forks are loaded and the boom is raised from the stowed position), which best describes the correct displacement of the combined center of gravity of the machine?

Combined center of gravity (CG) moves _____.

a) lower and towards the front of machine
b) higher and towards the front of machine
c) lower and towards rear of machine
d) higher and towards rear of machine

32. For the figure shown above (unleveled telehandler is raising a load while parallel to a slope), which statement is <u>not</u> true during operation?

a) The center of gravity is within the stability triangle
b) The center of gravity is on the left side of the centerline
c) The high boom angle may result in a tip-over
d) None of the above

33. The axle differential lock can be engaged while machine is moving.

a) TRUE
b) FALSE

34. Stabilizers _____.

a) help maintain machine stability during operation
b) should be used when overloading the telehandler
c) allow machine operation in gusty wind conditions
d) should be used while driving on rough terrain

35. A safe distance of at least _____ telehandler lengths must be maintained between other telehandlers.

a) one
b) two
c) three
d) depends on operator's best judgment

36. When the telehandler is loaded and ascending or descending a grade, the operator must always travel with the load _____.

a) sideways
b) uphill
c) downhill
d) either way

37. In the event of a tip-over, the operator should _____.

a) try to jump out of the cab
b) attempt to dump the load
c) shout for help
d) remain in the cab, grasp the steering wheel and lean away from the point of impact

38. Which statement **violates** safe and proper use of telehandlers?

Telehandlers should _____.

a) not be operated beyond the maximum allowable capacity
b) yield to pedestrians and emergency vehicles
c) not be operated if the operator's manual is missing
d) overtake a slower moving telehandler traveling in the same direction

39. When entering an elevator, enter _____ with load end _____. Once the telehandler has entered the elevator completely, set the brakes and follow proper shut-down procedures.

a) squarely, in front
b) squarely, trailing
c) slightly diagonally, in front
d) slightly diagonally, trailing

40. Select the incorrect statement.

a) Dock boards should have high friction surfaces and should be properly secured.
b) All dock boards should have proper markings that mention capacity rating.
c) Brakes and wheel chocks should be applied to trucks before being loaded with telehandlers.
d) Telehandler forks can be used to slowly open railroad car doors

41. The standard load chart is not applicable when lifting suspending loads.

a) TRUE
b) FALSE

42. Whenever a telehandler is intended to be used to lift and transport a suspended load, the operator must be trained and qualified in identifying and avoiding the potential hazards in handling a suspended load on the site.

a) TRUE
b) FALSE

43. (For operators receiving training in **U.S.A.** only)

As per OSHA 1910.178 (l) standards, operator retraining is required _____.

a) at intervals not exceeding 18 months
b) at intervals not exceeding 2 years
c) at intervals not exceeding 3 years
d) if the operator has been involved in an incident or observed to operate in an unsafe manner. Otherwise, only a re-evaluation of the operator is needed at a minimum of every three years.

44. (For operators receiving training in **Canada** only)

As per CSA B335 standards, operator retraining is required _____.

a) at intervals not exceeding 18 months
b) at intervals not exceeding one year
c) at intervals not exceeding two years
d) if the operator has been involved in an accident or observed to operate in an unsafe manner, or every three years, whichever comes first.

45. If at any time a powered industrial telehandler is found to be in need of repair, _____.

a) the operator must stop and repair the machine
b) the telehandler shall be tagged and removed from service until it is repaired by a qualified service technician
c) the telehandler can be parked after the job is complete
d) continue operating the machine

46. Scheduled maintenance inspections, above and beyond routine daily maintenance, should only be performed by qualified technicians.

a) TRUE
b) FALSE

47. Changing attachments does not affect the lifting capacity of the rough terrain forklift.

a) TRUE
b) FALSE

48. Prior to lifting a load with a telehandler, the operator should know _____.

a) the weight of the load
b) the platform load handling capability
c) the distance from the landing platform
d) the telehandlers lifting capability
e) All of the above

49. When picking up a load with forks you should:

a) Spread the forks as wide as possible
b) Make sure the load touches the back of the carriage
c) Carefully pick up the load, raise it to travel position and tilt the forks back slightly to cradle the load
d) All of the above

50. The first principle of safe operation is _____.

a) Perform a workplace inspection
b) Perform function tests
c) Avoid hazardous situations
d) Perform a pre-operation inspection
e) Only use the machine as intended

51. The pre-operation inspection includes ____.

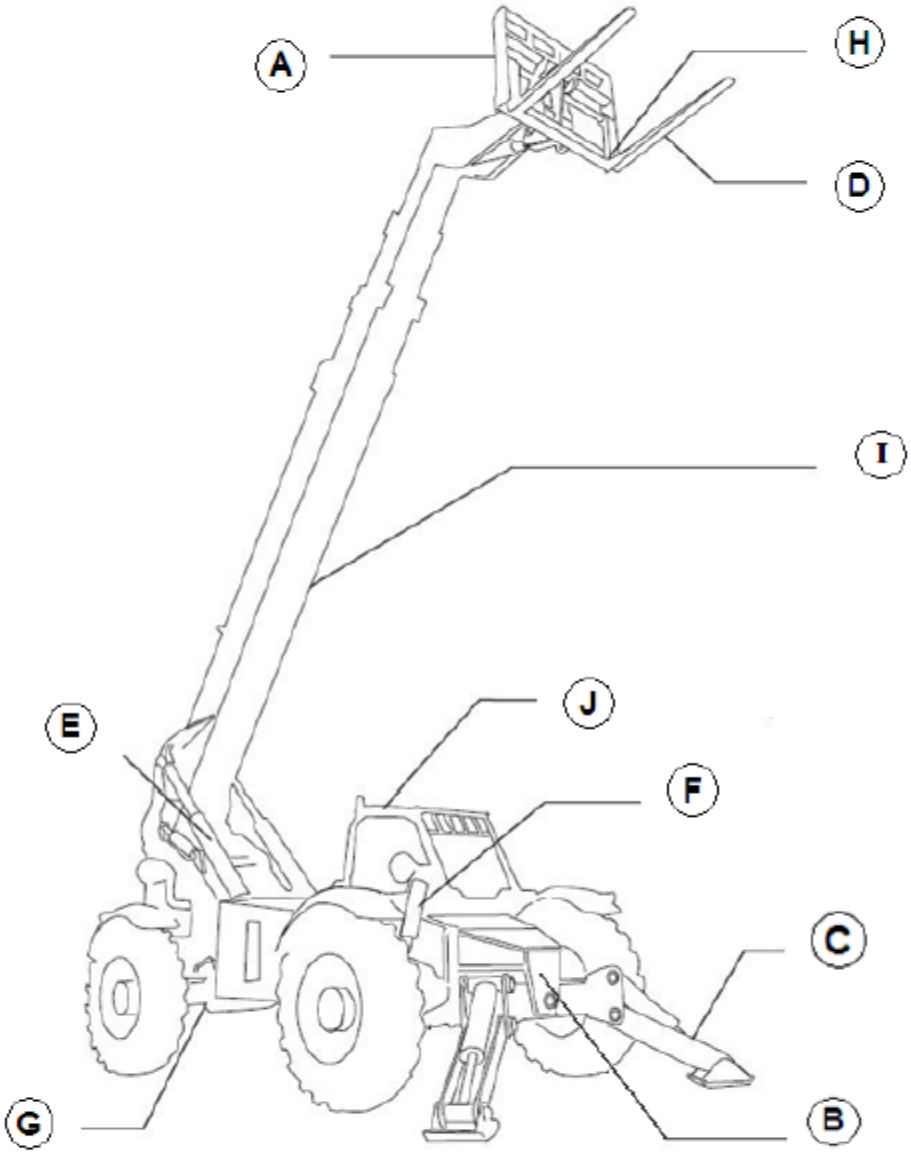
a) Testing the emergency stop
b) Looking for loose or missing parts
c) Driving the telehandler
d) Raising the boom to full extension

52. Function tests should be performed ____.

a) At the end of each shift
b) Prior to shutting the machine down
c) Prior to use
d) During breaks

53. **Telehandler Legend:** Correlate the labels with the correct components on the machine by filling in the blank space with the correct letter.

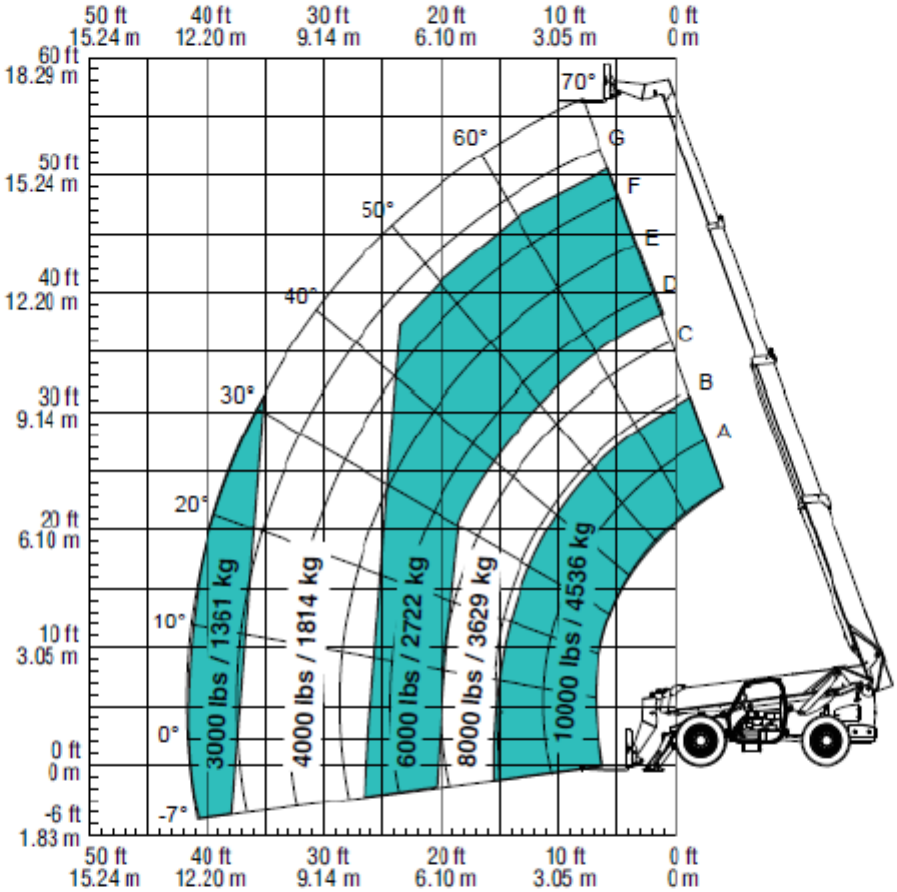
Note: These components may vary for different manufacturers. Refer to the course participant's guide.



1. Axle	_____	6. Boom	_____
2. Chassis	_____	7. Forks	_____
3. Frame leveling cylinder	_____	8. Fork stamp	_____
4. Stabilizers	_____	9. Fork carriage	_____
5. FOPS/ROPS Cab	_____	10. Lift Cylinder	_____

54. **Load charts:** Load charts are a critical part of machine operation. Select the correct option (stable or unstable) for each case by placing an "X" in the correct column.

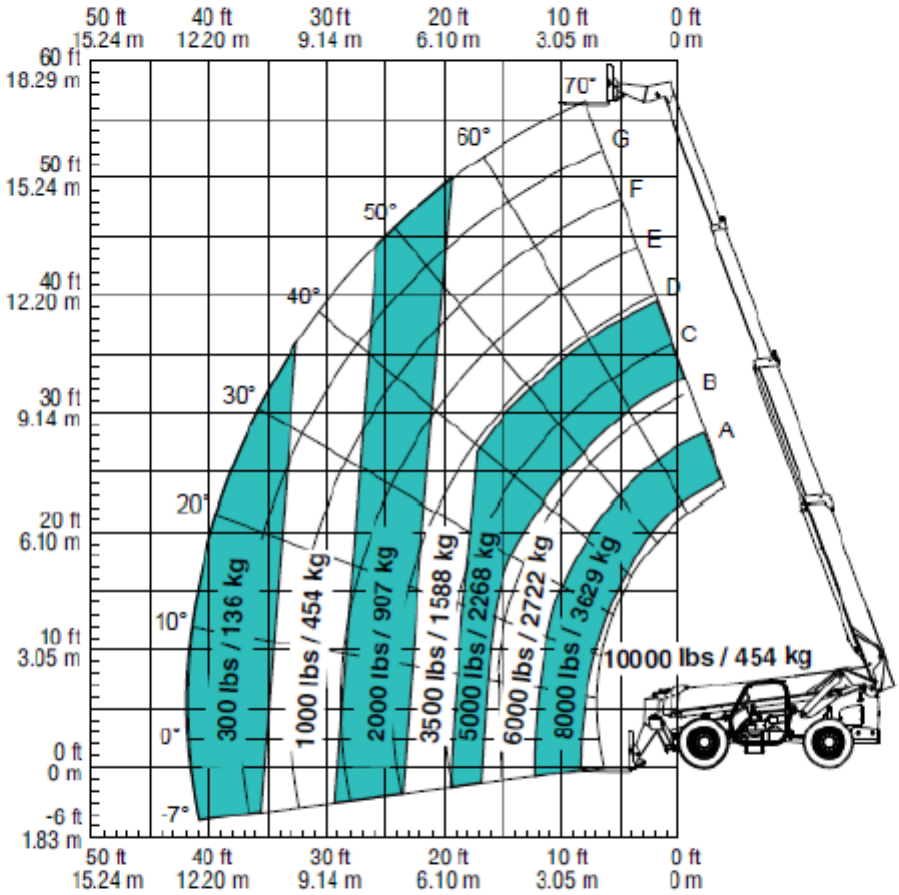
STABILIZERS DOWN



Case	Load weight	Horizontal extension	Boom Angle	Stable	Unstable
1	10,000 lbs	20 ft	40°		
2	3,000 lbs	40 ft	8°		
3	7,000 lbs	10 ft	10°		
4	5,000 lbs	28 ft	5°		
5	8,000 lbs	30 ft	10°		
6	4,000 lbs	38 ft	20°		

55. **Load charts:** Load charts are a critical part of machine operation. Select the correct option (stable or unstable) for each case by placing an "X" in the correct column.

STABILIZERS UP



Case	Load weight	Horizontal extension	Vertical extension	Stable	Unstable
1	3,000 lbs	20 ft	40°		
2	1,000 lbs	40 ft	3°		
3	5,000 lbs	15 ft	33°		
4	3,500 lbs	25 ft	2°		
5	8,000 lbs	10 ft	5°		
6	4,000 lbs	20 ft	35°		

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